



École Polytechnique Fédérale de Lausanne

Design, Simulation, and Tooling Work in Integrated Photonics
at the Laboratory of Photonics and Quantum Measurements

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Internship Report

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I thank Alessio Zicoschi for his guidance on the three software tools described in this report, which gave me the opportunity to explore recent AI tools and build a tool end-to-end for the group. I am also grateful to Marco Golino and Shreya Shah, together with Alessio, for our collaboration on the PowerPoint semantic search tool, a joint effort among the four of us from which I learned a great deal. We hope these tools will continue to support the group and its lab infrastructure well beyond our time here.

I was fortunate to witness Prof. Kippenberg's Marcel Benoist Prize lecture, attend the INGEN conference, and observe the public defences of several group members, experiences that, in a short time, gave me a real taste of what pursuing a PhD involves.

Being part of such a large group also taught me valuable lessons in teamwork, organisation, and leadership. I would like to thank the wider LPQM group for a welcoming and stimulating environment. Several members have become close friends, and their warmth and kindness made me feel at home despite being far from mine. The lunch table conversations, in particular, were a source of both enjoyment and learning, and I came away from them having learned a great deal.

Abstract

This report documents work carried out during a research internship (7 May to 31 July 2026) in the Laboratory of Photonics and Quantum Measurements (LPQM) at the École Polytechnique Fédérale de Lausanne (EPFL), under the supervision of Prof. Tobias J. Kippenberg and the guidance of Xinru Ji. Two design and modeling projects form the technical core of the internship: the design, simulation, and mask layout of a chirped-Bragg-grating Fabry-Perot microresonator in silicon nitride for dissipative Kerr soliton generation, and the implementation and validation of a roughness-limited scattering-loss model for silicon nitride waveguides against a published high-confinement photonic platform. Alongside this design work, three technical notes were contributed to the laboratory's internal documentation system, covering its PowerPoint slide-search tool, the group's GDS mask-layout library, and its Lumerical LSF scripting workflow, and three internal software tools were built to support the group's daily operations: a semantic search interface for the lab's slide archive, a searchable catalog for the technical-notes collection, and a browser-based front end for the fabrication design-file share. This report presents the two design projects in full, points to the three technical notes contributed during the internship, and summarizes the three software tools.

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Chapter 1

Introduction

The Laboratory of Photonics and Quantum Measurements (LPQM), led by Prof. Tobias J. Kippenberg at the École Polytechnique Fédérale de Lausanne (EPFL), works on integrated photonic devices spanning dissipative Kerr soliton (DKS) microcombs, cavity optomechanics, and superconducting quantum circuits, with a shared reliance on silicon nitride (Si_3N_4) photonic integrated circuits (PICs) as a low-loss, wafer-scale fabrication platform. Two design and simulation projects carried out during this internship sit within that platform: the design of a Fabry-Perot (FP) microresonator whose dispersion is set by its mirrors rather than by its waveguide, targeted at soliton microcomb generation, and the implementation and validation of a model for the scattering loss that residual waveguide roughness imposes on any Si_3N_4 resonator.

Around this design work, the internship also produced three contributions to the laboratory's technical notes and three small software tools built to support the group's day-to-day operations: searching an archive of presentation slides, browsing the technical-notes collection itself, and looking up the files associated with a fabrication design.

The remainder of this report is organized as follows. Chapter 2 presents the Fabry-Perot microresonator project: the cavity and soliton model, the validated symmetric and asymmetric cavity designs, their fabrication tolerance, and the resulting fabrication mask. Chapter 3 presents the scattering-loss modeling project: the theoretical model, its validation against a published high-confinement Si_3N_4 platform, its extension to 1 μm operation, and a benchmarking analysis of the sidewall scattering contribution against measurement. Chapter 4 points to three technical notes contributed to the laboratory's documentation during the internship, covering its PowerPoint slide-search tool, the group's GDS mask-layout library, and its Lumerical scripting workflow. Chapter 5 summarizes three software tools built for the group. Chapter 6 concludes.

Chapter 2

Dispersion-Engineered Chirped-Bragg-Grating Fabry-Perot Microresonators for Dissipative Kerr Soliton Generation in Silicon Nitride

Abstract

Dissipative Kerr solitons (DKS) in integrated microresonators generate broadband, mutually coherent frequency combs from a single continuous-wave pump and are of interest as compact comb sources for coherent communications, ranging, and spectroscopy. Almost all integrated DKS sources have been realized in ring resonators, where the cavity dispersion is fixed by the waveguide cross section and offers limited independent tuning. A linear Fabry-Perot (FP) cavity, in which the two end mirrors themselves are engineered to set the dispersion, removes this constraint and additionally avoids the unintentional mode splitting that affects ring resonators. This work reports the design, simulation, and mask layout of a chirped-Bragg-grating (CBG) FP microresonator in a thin silicon nitride (SiN) waveguide platform. A coupled-mode cavity model and a standing-wave Lugiato-Lefever equation (LLE) solver, including thermal feedback, were used to keep the second-order dispersion D_2 below 10 MHz, a regime that favors robust single-soliton formation, and to identify four cavity geometries, differing in mirror length and grating contrast, that support a stable soliton with loaded quality factors of 7×10^6 to 1.03×10^7 and $D_2/2\pi$ between 8.0 and 9.1 MHz at on-chip pump powers of 220 to 260 mW. All four designs tolerate realistic lithographic width error (D_2 shifts by less than 3% and Q by 1 to 28% over a ± 50 nm range). An asymmetric variant of the cavity, with one mirror made essentially fully reflective and the other weakly transmissive, was further designed to provide a dedicated soliton output port while keeping the loaded Q close to that of the symmetric designs;

two such devices, with drop-port transmissions of 0.12% and 0.25%, were confirmed by simulation to support a soliton over on-chip pump powers of 250 to 400 mW. All designs have been laid out and verified on a single fabrication mask, which is now ready for fabrication and experimental characterization.

2.1 Introduction

A dissipative Kerr soliton (DKS) is a self-sustaining, localized pulse of light that circulates in a driven, high-quality-factor optical resonator once the resonator’s chromatic dispersion, its intrinsic loss, and the Kerr nonlinearity of its material balance against each other. Pumping such a cavity with a single continuous-wave laser then produces, at the output, a broadband comb of optical frequencies, mutually coherent and phase-locked to the pump. Integrated DKS microcombs are of interest because a single chip and a single laser can replace the mode-locked laser systems conventionally used for frequency-comb generation, with applications in coherent optical communications, LIDAR ranging, dual-comb spectroscopy, microwave photonics, and calibration of astronomical spectrographs [3,4].

As Wildi et al. [1] note, prior to their demonstration integrated DKS sources had been pursued exclusively in traveling-wave ring resonators, where the dispersion is set by the waveguide cross section and offers only limited independent tuning once a material platform and film thickness are fixed. Rings additionally support two frequency-degenerate counter-propagating modes; fabrication imperfections couple these modes together at random locations set by the specific defects of the specific chip, producing unintentional avoided mode crossings that locally distort the dispersion and can suppress soliton formation in a way that is not part of the design. Wildi et al. [1] demonstrated, for the first time, continuous-wave-driven DKS generation in a wafer-fabricated, chip-integrated Fabry-Perot (FP) microresonator whose broadband dispersion is dominated by its mirrors rather than by the waveguide, using photonic-crystal reflectors. Because an FP cavity’s two counter-propagating fields are already present by construction at every mirror reflection, rather than arising from an accidental coupling between two otherwise independent modes, this geometry was shown to be free of the unintentional mode splitting seen in ring devices fabricated on the same wafer.

Chirped-Bragg-grating (CBG) FP mirrors extend this approach by giving the mirror itself a spatially varying reflection period, so that the mirror’s group delay dispersion (GDD), and hence the cavity’s D_2 , becomes an explicit, tunable design parameter rather than an incidental property of a fixed reflection band. Two studies illustrate how strongly this parameter can be varied in practice: Wildi et al. [1] used an unchirped mirror and obtained a small anomalous dispersion, $D_2/2\pi = 210$ kHz, sufficient for a first demonstration but not itself a design target; Cheng et al. [2] used a chirped, apodized CBG mirror deliberately tuned for a large dispersion, $D_2/2\pi = 90$ MHz, more than two orders of magnitude larger, trading comb bandwidth for higher power per comb line suited to optical-communication applications. Both works note the same underlying trade-off: increasing D_2 raises the intracavity comb-line power at the cost of the comb’s optical bandwidth.

This work reports the design of a CBG FP DKS microresonator, built in a thin silicon nitride (SiN) waveguide platform (200 nm core thickness). The motivation is an ongoing effort to generate a high-power, low-noise soliton comb source whose individual lines are subsequently amplified and channelized for parallel ranging and communication experiments, which places a premium on a comb with enough per-line power to survive downstream demultiplexing and modulation while retaining enough lines to be useful as a multi-channel source. Because a thin, weakly confined SiN waveguide differs from the thicker, more strongly confining platforms used in references [1,2], the cavity, mirror, and soliton dynamics were modeled from first principles rather than adapted directly from either prior report. As discussed in Subsection 2.3.1, keeping D_2 small, while staying above a floor set by modulation instability, favors robust single-soliton formation, and this reasoning set the dispersion target for the design. The same design and simulation approach was subsequently extended to an asymmetric cavity variant in which one mirror is made essentially fully reflective and the other weakly transmissive, giving the soliton a dedicated output port separate from the pump input.

2.2 Cavity design principle and soliton dynamics

2.2.1 Dispersion and the resonance ladder

The frequency of resonance mode μ ($\mu = 0$ denotes the pumped mode) can be written as $\omega(\mu)$, and its deviation from a perfectly evenly spaced ladder defines the integrated dispersion,

$$D_{\text{int}}(\mu) = \omega(\mu) - \omega(0) - \mu D_1 = \frac{D_2}{2} \mu^2 + \frac{D_3}{6} \mu^3 + \dots$$

D_1 is 2π times the free spectral range (FSR), and D_2 is the second-order (quadratic) term that determines whether the cavity supports bright solitons. Positive D_2 (anomalous dispersion) balances the Kerr-induced self-steepening of an intense pulse and allows a stationary sech-squared soliton to exist; negative D_2 (normal dispersion) instead supports dark-pulse structures.

In a ring, D_1 is set by the round-trip length and D_2 by the waveguide's own group-velocity dispersion. In an FP cavity built from two Bragg-grating mirrors and a connecting spacer waveguide, a chirped grating adds a wavelength-dependent penetration depth: light near one edge of the reflection band turns around close to the grating's entrance, while light near the other edge penetrates further before reflecting. This wavelength-dependent effective mirror position is exactly what a mirror's group delay dispersion (GDD) describes, the same quantity used for chirped-mirror design in mode-locked lasers. Writing the round-trip phase as $\phi_{\text{rt}}(\omega)$, its Taylor expansion about the pump frequency ω_0 ,

$$\phi_{\text{rt}}(\omega_0 + \delta\omega) = \phi_{\text{rt}}(\omega_0) + \tau_{\text{rt}} \delta\omega + \frac{1}{2} \text{GDD}_{\text{rt}} \delta\omega^2 + \dots, \quad \tau_{\text{rt}} = \frac{d\phi_{\text{rt}}}{d\omega}, \quad \text{GDD}_{\text{rt}} = \frac{d^2\phi_{\text{rt}}}{d\omega^2},$$

gives, order by order through the resonance condition $\phi_{\text{rt}}(\omega_\mu) = 2\pi\mu$, the round-trip time (hence D_1) at first order and

$$D_2 = -\frac{D_1^3}{2\pi} \text{GDD}_{\text{rt}}$$

at second order, a relation confirmed against the D_2 and GDD values reported in references [1,2]. Because GDD scales with mirror length and, for a linear chirp, inversely with the chirp rate, while D_1 is set largely by the spacer length, D_2 can be engineered by the mirror geometry with only weak coupling to the spacer waveguide's own cross section.

2.2.2 Mirror and cavity model

Each mirror was modeled as a chirped grating built from Bragg unit cells, with the period $\Lambda(z)$ swept linearly along the mirror length and the tooth-width contrast windowed by a flat-top hyperbolic-tangent apodization envelope (flat over a central fraction of the mirror, rolling off smoothly at both ends) to suppress ripple in the resulting dispersion. For a rectangular corrugation of duty cycle D , the coupling coefficient is

$$\kappa = \frac{2\Delta n \sin(\pi D)}{\lambda_B}, \quad \lambda_B = 2\bar{n}\Lambda, \quad \bar{n} = Dn_1 + (1-D)n_2,$$

maximized at $D = 0.5$. Each mirror was cascaded cell by cell as a coupled-mode transfer matrix, combined with the spacer propagation matrix to give the full round-trip phase $\phi_{\text{rt}}(\omega)$, and the resonance ladder obtained by locating the frequencies at which $\phi_{\text{rt}} = 2\pi m$. A cubic fit to this ladder gives D_1 , D_2 , and D_3 directly, together with the mirror reflectivity and the loaded quality factor Q .

At the input mirror, the field reflected back toward the bus waveguide is the direct mirror reflection combined with the field that leaks back out after circulating around the cavity; the two interfere destructively, and essentially all incident power is coupled into the cavity, when the two mirror reflectivities and the round-trip loss and phase satisfy $|r_1| = |r_2|a^2$ and $\arg(r_1) = \arg(r_2) + \arg(a^2)$, with r_1, r_2 the mirror amplitude reflection coefficients and a the round-trip field amplitude retained. This is the FP analogue of critical coupling and guided the choice of mirror reflectivities for both the symmetric cavities (Section 2.3) and the asymmetric, drop-port cavity (Section 2.4), where an essentially unity front-mirror reflectivity additionally suppresses back-reflection of the pump toward the source. In the small-loss limit, the loaded quality factor reduces to the standard cavity-theory result $Q \approx \omega_0 \tau_{\text{rt}} / (T_1 + T_2 + 2\alpha L_{\text{cav}})$, with T_1, T_2 the mirror power transmissions and α the spacer's power propagation loss; the full transfer-matrix calculation used throughout this work reduces to this expression in the appropriate limit.

2.2.3 Soliton dynamics in a standing-wave cavity

Soliton formation was modeled with the Lugiato-Lefever equation (LLE) [5], adapted for the standing-wave character of an FP cavity following references [1,6]:

$$\frac{\partial \psi}{\partial \tau} = -(1 + i\zeta)\psi + i \frac{d_2}{2} \frac{\partial^2 \psi}{\partial \theta^2} + i|\psi|^2\psi + 2i\langle |\psi|^2 \rangle \psi + F,$$

where $\psi(\theta, \tau)$ is the normalized intracavity field, θ the round-trip coordinate, τ the normalized (slow) time, and

$$d_2 = \frac{2D_2}{\kappa}, \quad \zeta = \frac{2\Delta}{\kappa}, \quad F^2 = \frac{8\eta_{\text{in}} g_0 P_{\text{in}}}{\kappa^2 \hbar \omega_0},$$

with $\kappa = \omega_0/Q$ the cavity linewidth, $\eta_{\text{in}} = 0.10$ the fraction of total loss due to the input coupler, and g_0 the single-photon Kerr shift. The fourth term on the right of the field equation, absent in a traveling-wave ring cavity, arises because the two counter-propagating fields present in a standing-wave cavity cross-phase-modulate one another in addition to self-phase-modulating themselves; for a spatially uniform field this makes the net Kerr response three times as strong as in a ring, $c = 3$ rather than $c = 1$ in the corresponding continuous-wave bistability relation $c^2 S^3 - 2c\zeta S^2 + (1 + \zeta^2)S = F^2$. Because the extra term depends only on the total circulating power and not on mode index, it can be absorbed into a redefinition of the effective detuning, so that soliton theory developed for ring cavities applies to the FP case once this factor of three, and thermal detuning drift, are included [1]. A slower thermal state Θ , tracking absorbed pump power with relaxation rate m_1 and coupling strength m_2 ,

$$\frac{d\Theta}{d\tau} = m_1(m_2\langle |\psi|^2 \rangle - \Theta), \quad \zeta_{\text{eff}} = \zeta - \Theta,$$

shifts the effective detuning seen by the field, and both the idealized and the thermally coupled dynamics were evaluated for every design below.

The field equation was integrated with a split-step method (the linear terms applied exactly in Fourier space, the nonlinear and pump terms with a fourth-order Runge-Kutta step) over an adiabatic sweep of the pump detuning; a soliton state was identified from the resulting energy-versus-detuning trace as a flat, low-variance plateau following the characteristic drop from the modulation-instability regime, and its temporal profile fit to a sech-squared envelope to extract pulse duration and spectral bandwidth.

2.3 Symmetric cavity design

2.3.1 Design constraints

A smaller D_2 makes soliton formation easier when taken in isolation, holding the cavity's linewidth and nonlinear drive fixed: the soliton-supporting range of detuning widens and less pump power is

needed to sustain it, so the pump-power budget available for this work favored keeping D_2 as low as practical. An initial search over cavity spacer length, mirror chirp fraction, mirror length, and grating tooth-width contrast, subject to a mirror reflectivity $R(\lambda_0) \geq 0.999$ and loaded $Q \geq 5 \times 10^6$, showed that this cannot be pushed indefinitely: minimizing D_2 without a lower bound drives the design toward $D_2/2\pi$ of order 0.7 MHz, at which point the soliton solver produces only modulation instability rather than a clean soliton. Reaching that small a D_2 requires a substantially longer spacer and a larger chirp fraction than the designs eventually adopted, and because the single-photon Kerr shift that sets the pump strength available at fixed on-chip power scales inversely with the cavity round-trip length, the same geometry change that suppresses D_2 also suppresses this available drive. Considered together, rather than each in isolation, the two effects leave too little pump strength to sustain a soliton against the modulation-instability gain, which itself broadens as D_2 shrinks. A lower bound was therefore imposed on D_2 , and the design was kept below $D_2/2\pi = 10$ MHz throughout, giving the band used for the rest of this work.

The choice of spacer waveguide width also involves a trade-off between nonlinearity and dispersion sensitivity. A narrower spacer confines the mode more tightly, reducing the effective mode area A_{eff} and correspondingly increasing the Kerr nonlinearity per unit power, but its group-velocity dispersion β_2 is then larger in magnitude and more sensitive to width variation. A wider spacer (of order 2.5 to 3.5 micrometers for the waveguide thickness used here) gives a weaker per-watt nonlinearity but a β_2 that has largely saturated with width and is correspondingly more tolerant of fabrication variation; this second regime was adopted throughout, since the resulting cavities are pumped comfortably above the modulation-instability threshold (Subsection 2.3.3) and benefit more from dispersion robustness than from maximal nonlinearity.

2.3.2 Design search and validated candidates

Within this D_2 band, a multi-dimensional search over spacer length, mirror length, chirp fraction, spacer width, and grating tooth widths, with A_{eff} , group index n_g , and the spacer's own β_2 and β_3 obtained from a finite-difference eigenmode simulation of the actual waveguide cross section, identified a common cavity backbone (2 mm spacer, 2.5 micrometer spacer width, chirp fraction 0.30) and, within it, four mirror geometries carried through full soliton-dynamics validation (Table 1). All four support a stable soliton state, confirmed by direct simulation of both the idealized and thermally coupled dynamics; depending on device and branch this is a single circulating pulse or, as for the reference device below, two pulses at fixed relative positions (a simple soliton crystal).

Device 3 is used as the reference design in the remainder of this report: it shows the widest confirmed detuning range over which the soliton is stable and is reachable by a direct adiabatic scan on both the idealized and thermally coupled branches, whereas Device 4 requires the simulation to be seeded from Device 3's converged state to locate the same soliton branch. The three loss-independent cavity parameters shared by all four devices (obtained from the same finite-difference

Table 1: Validated symmetric-cavity designs.

Device	Mirror length	Tooth widths (narrow/wide)	Loaded Q	$D_2/2\pi$	FSR
1	800 μm	0.85 / 1.85 μm	7.0×10^6	8.16 MHz	31.7 GHz
2	1600 μm	0.85 / 1.60 μm	1.02×10^7	9.05 MHz	25.7 GHz
3 (reference)	1600 μm	1.00 / 2.50 μm	1.03×10^7	9.07 MHz	25.6 GHz
4	1600 μm	1.15 / 2.65 μm	1.02×10^7	9.02 MHz	25.5 GHz

Confirmed soliton operating point: Device 1, $P_{\text{in}} = 259$ mW, $F^2 = 20$; Device 2, $P_{\text{in}} = 226$ mW, $F^2 = 30$; Device 3, $P_{\text{in}} = 223$ mW, $F^2 = 30$; Device 4, confirmed via seeded scan from Device 3.

eigenmode calculation at the 2.5 micrometer spacer width) are $A_{\text{eff}} = 1.77 \mu\text{m}^2$, $n_g = 1.805$, and $\beta_2 = +1050 \text{ ps}^2/\text{km}$.

Figure 1 shows the chirped-mirror period profile used for the reference device, and Figure 2 the resulting integrated dispersion, a clean parabola confirming a dominant, well-controlled second-order term with $D_2/2\pi = 9.068$ MHz and negligible visible ripple from higher orders over the fitted range.

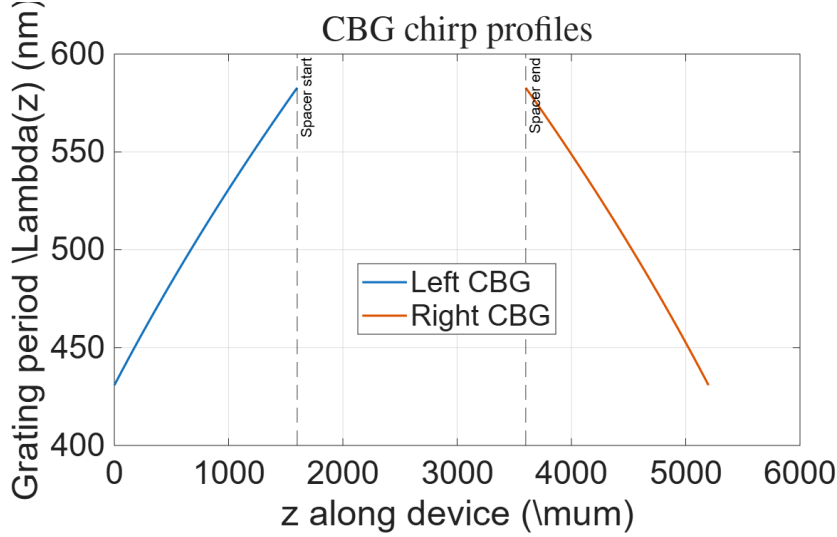


Figure 1: Grating period $\Lambda(z)$ along each mirror of the reference device (Device 3), linearly chirped and mirrored about the spacer so that the long-period, red-shifted edge of both gratings faces the spacer.

For the reference design, the confirmed soliton state has a plateau spanning roughly $\Delta\zeta = 0.8$ in normalized detuning and a comb visible over approximately 100 to 120 longitudinal modes above the numerical noise floor. Figure 3 shows the intracavity field intensity across the full detuning scan (thermally coupled branch): two stationary intensity peaks, at fixed positions in θ , persist together over a finite range of ζ following an initial modulation-instability region and preceding the collapse to the low-power background, indicating a two-soliton state rather than a single circulating

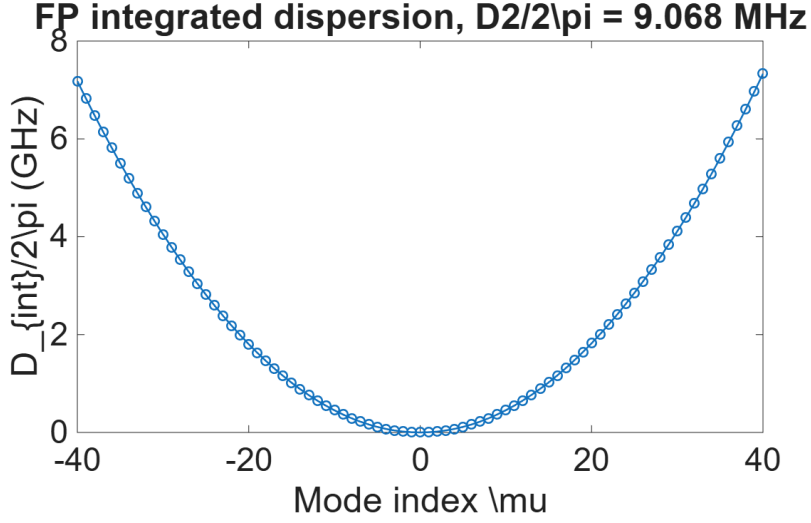


Figure 2: Integrated dispersion $D_{\text{int}}(\mu)$ of the reference device, obtained from the cubic fit to the simulated resonance ladder. The parabolic shape confirms anomalous, dominantly second-order dispersion, $D_2/2\pi = 9.068$ MHz.

pulse. Figure 4 shows the corresponding energy-versus-detuning trace, and Figure 5 the comb spectrum extracted at the operating point: a smooth envelope on a logarithmic scale with a fine ripple superimposed, the interference signature of the two coexisting solitons.

A direct time-domain sech-squared characterization was also carried out for the reference device: at a single-soliton point within the same detuning plateau (the plateau also contains the two-soliton state of Figure 3), the fit gives a 1356 fs pulse (fit quality 0.998) and a comb spanning 59 nm (Figure 6), comparable to the 856 fs, 55 nm obtained for the first exploratory design point of this project ($D_2/2\pi = 8.5$ MHz, $Q = 7.0 \times 10^6$).

The equivalent integrated-dispersion curves for Devices 1, 2, and 4 are given in Supplementary Figures S1 to S3.

2.3.3 Fabrication tolerance

Sweeping the fabricated waveguide width by ± 50 nm at fixed tooth-width contrast, representative of realistic lithographic error, shifts D_2 by less than 3% for all four devices in Table 1, but the corresponding shift in Q ranges from about 1% (Device 3) to about 28% (Device 1). Among these four devices, the spread tracks mirror length more closely than the grating's tooth-width contrast: Device 2, which has the smallest tooth contrast of the four, nonetheless shows one of the smallest Q variations, while Device 1, with the shortest mirror, shows by far the largest. Mirrors of 1200 micrometers or longer were more tolerant of fabrication width error than the 800-micrometer mirror

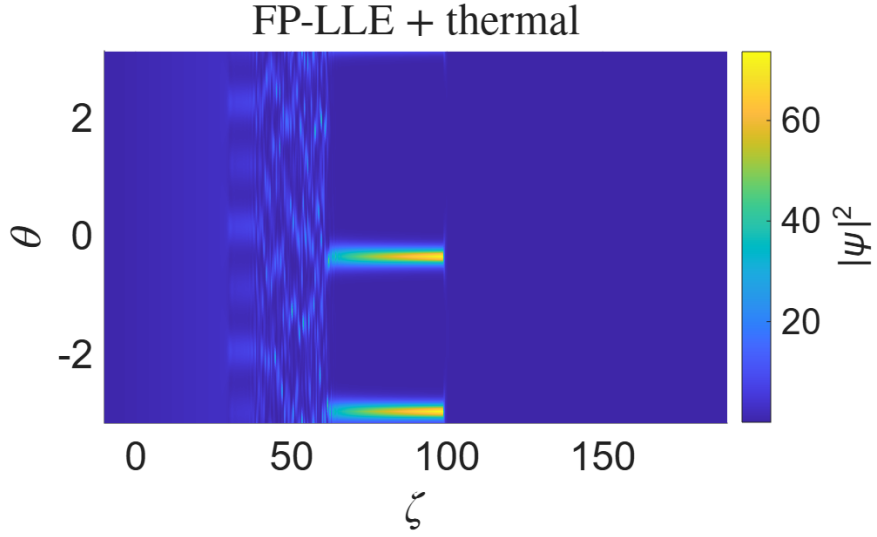


Figure 3: Intracavity field intensity $|\psi(\theta, \zeta)|^2$ of the reference device across the detuning scan (thermally coupled branch). Two stationary peaks in θ persist together over a finite ζ range, indicating a two-soliton state.

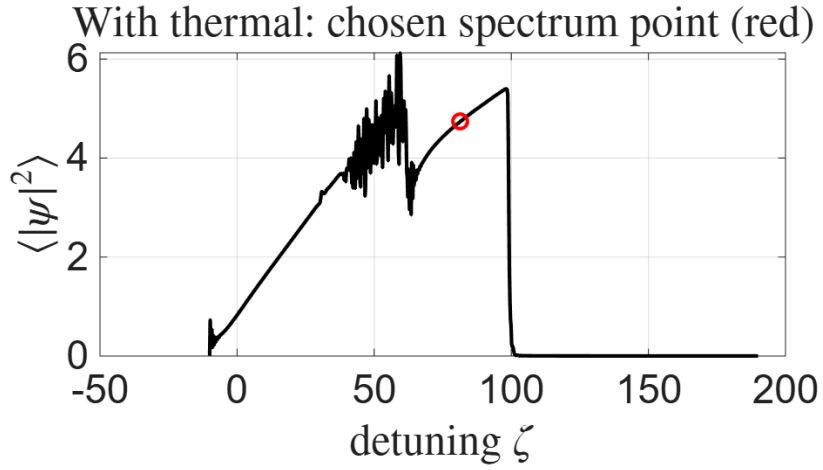


Figure 4: Mean intracavity intensity $\langle |\psi|^2 \rangle$ versus normalized detuning ζ for the reference device (thermally coupled branch), showing the modulation-instability region followed by the soliton step used for characterization.

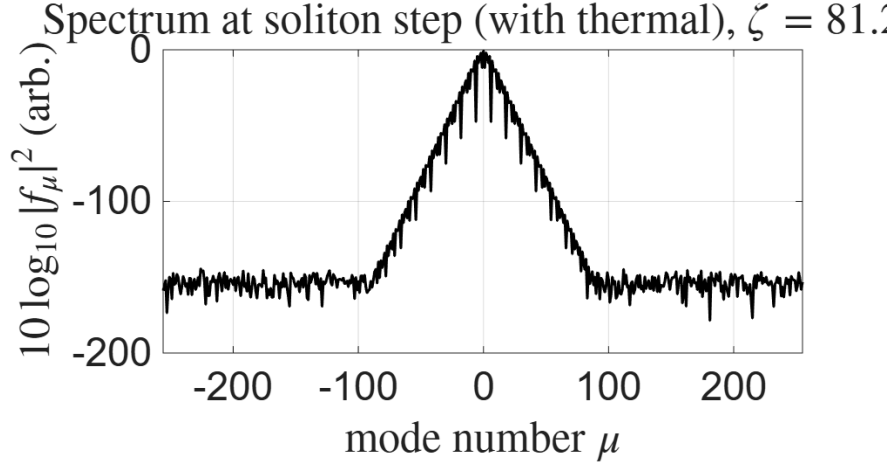


Figure 5: Comb spectrum $10 \log_{10} |f_{\mu}|^2$ at the characterized soliton operating point. The fine ripple on the envelope is the interference pattern between the two solitons seen in Figure 3.

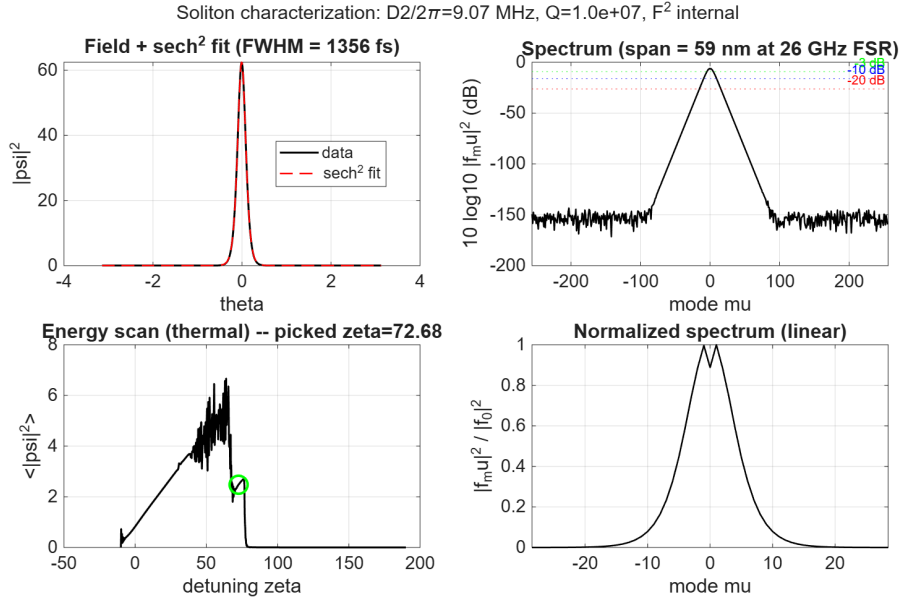


Figure 6: Time-domain sech-squared characterization of the reference device (Device 3) at a single-soliton point within the same detuning plateau as Figure 3 to Figure 5 ($\zeta = 72.7$): intracavity field with the fitted sech-squared envelope (top left), comb spectrum with the $-3/-10/-20$ dB bandwidths marked (top right), the energy-versus-detuning scan with the characterized point marked (bottom left), and the normalized linear spectrum (bottom right).

across this set (Figure 7, Supplementary Table 1).

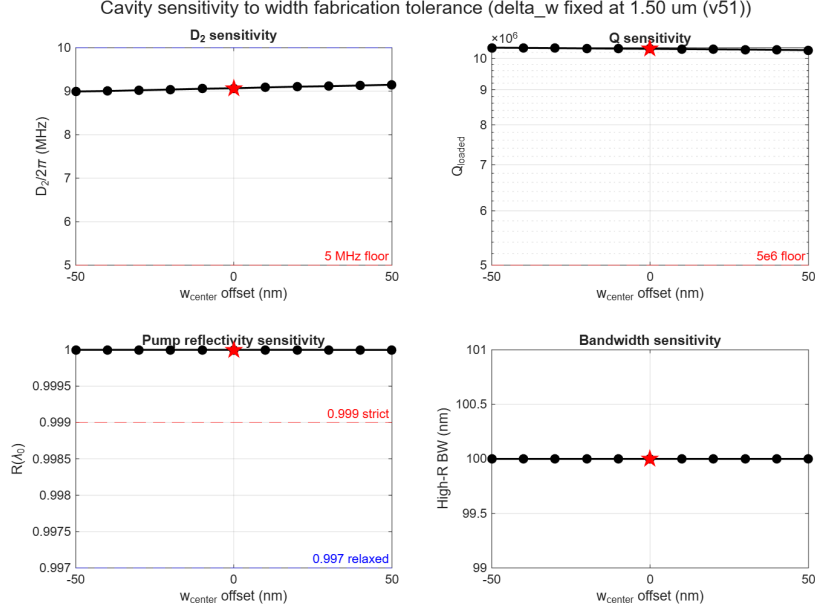


Figure 7: Sensitivity of the reference device's D_2 , loaded Q , pump reflectivity, and high-reflectivity bandwidth to a common ± 50 nm shift of the fabricated waveguide width (tooth-width contrast held fixed). All four metrics stay within the required feasibility bounds (dashed lines) across the full range.

2.3.4 Optical verification of the taper and the bus coupling

The cavity model of Section 2.3 (mirror and cavity model) treats the mirror-to-spacer junction as an abrupt width step and does not include the bus waveguide used to couple pump light into, and comb light out of, a fabricated device. Both were checked separately with electromagnetic simulation of the actual waveguide geometry before being committed to the mask, in two independent simulations reported in turn below: the taper in Lumerical MODE Solutions, and the bus-to-cavity coupling in Lumerical FDTD Solutions.

Taper insertion loss. The 100 μm linear taper bridging the apodized grating end (1.2 μm wide) to the spacer waveguide (2.5 to 3.5 μm wide, depending on device) was simulated with a three-dimensional eigenmode expansion (EME) calculation of the fundamental TE mode's transmission through the taper, on the 200 nm SiN cross section used throughout this work. At 1550 nm the simulated insertion loss is 0.0026 dB, and it stays between 0.0018 and 0.0034 dB across the 1500 to 1600 nm band. The taper is therefore confirmed adiabatic as drawn, with TE-to-TE coupling effectively lossless; the same conclusion applies to all four devices in Table 1, whose actual width step

(2.5 to 1.0-1.15 μm) is smaller than, and consequently easier than, the 3.5 to 1.2 μm case simulated here.

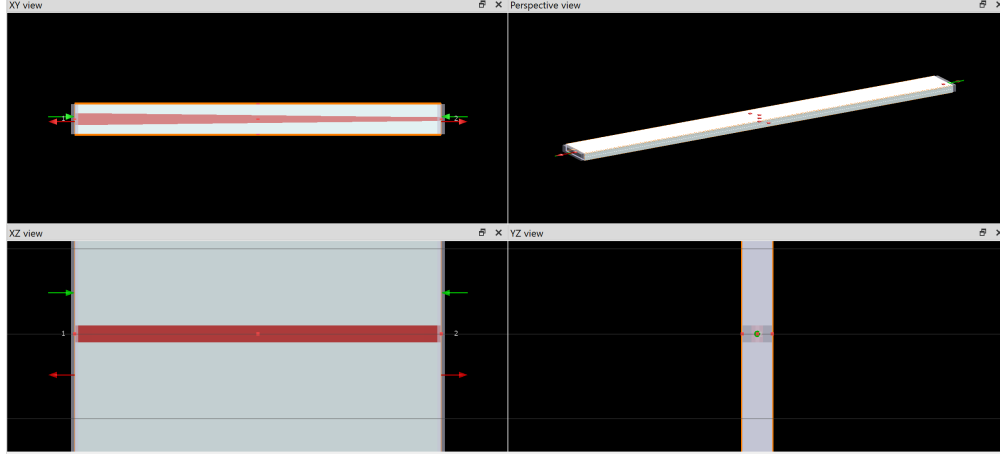


Figure 8: Lumerical MODE Solutions EME simulation window for the taper insertion-loss calculation: the taper geometry in cross-section (XY, XZ, YZ views) and perspective, with the input and output ports (1 and 2) used to compute the transmission reported in the text.

Bus-to-cavity coupling. The coupling rate between the bus waveguide and the cavity is calculated as a function of the bus-cavity gap with a three-dimensional finite-difference time-domain (FDTD) simulation of the two waveguides side by side across the gap, including the Euler S-bends that bring the bus from the edge-coupler line to the coupling gap and back out again (Figure 13), so that any coupling contributed by the bends themselves is captured directly in the same simulation rather than estimated separately. For each gap, the coupled section is simulated at several straight-section lengths, and the total accumulated coupling phase is fit linearly against length: the slope gives the coupling rate per unit length of the straight section, and the intercept gives the fixed contribution of the two bends. The resulting coupling rate is converted to an external coupling rate $\kappa_{\text{ex}}(\text{gap})$ and compared against each device's own intrinsic loss rate $\kappa_{\text{int}} = \omega_0/Q$ from Table 1; the gap at which $\kappa_{\text{ex}} = \kappa_{\text{int}}$ is the critical-coupling point, where the bus couples exactly as much power out of the cavity as the cavity loses to intrinsic loss. This simulation sweep, covering all devices in Tables 1 and 2, was still running at the time of writing.

2.4 Asymmetric cavity for a dedicated soliton output port

The symmetric devices of Section 2.3 couple pump and comb through the same bus waveguide. A cavity in which one mirror is made essentially fully reflective at the pump wavelength and the other given a small, deliberate transmission provides a dedicated output port on the second mirror, separating the extracted soliton comb from the residual pump and reducing back-reflection toward the pump laser (see the mirror and cavity model above).

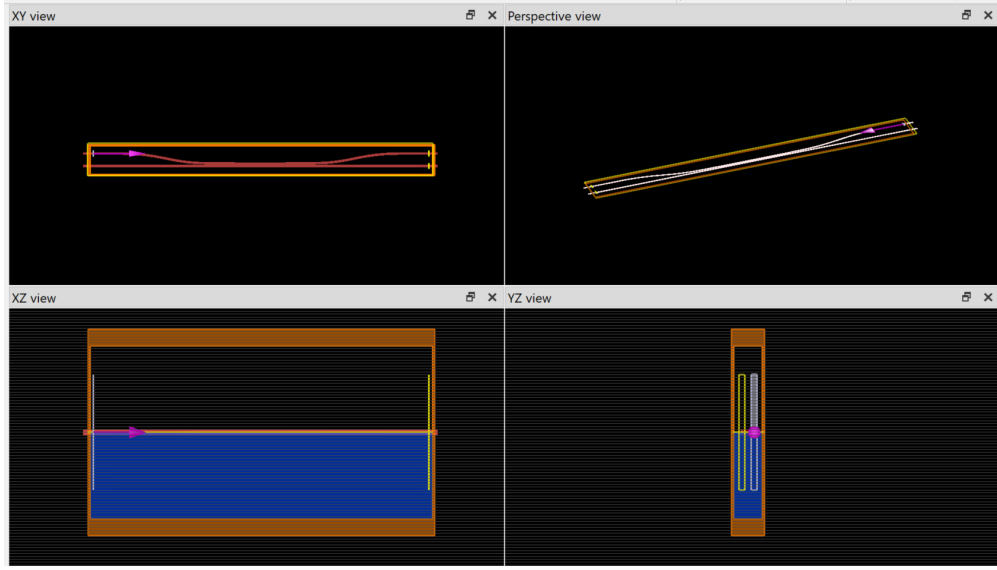


Figure 9: Lumerical FDTD simulation window for the bus-to-cavity coupling calculation: the bus waveguide, including its Euler S-bends, and the cavity spacer waveguide side by side across the coupling gap, used to extract the external coupling rate $\kappa_{\text{ex}}(\text{gap})$.

The design aim was to keep the loaded Q of this cavity close to the value validated for the symmetric devices, since any mirror transmission is itself a loss channel and adding one only works if it stays small next to the intrinsic propagation loss that otherwise sets Q . A first attempt sized the second mirror's transmission for a comfortably detectable output signal (about 1%), which brought Q down to about 3×10^6 , well short of the 7 to 8×10^6 of the symmetric devices; matching Q required engineering the transmission down to a few tenths of a percent instead (Supplementary Table 2). Because this cavity's Q is lower than the symmetric devices', the design target used was the dispersion normalized to the cavity linewidth, $d_2 = 2D_2/\kappa$, rather than D_2 in absolute units.

Two asymmetric geometries with independently chosen mirror lengths were confirmed by simulation to support a stable soliton (Table 2).

Table 2: Validated asymmetric drop-port designs.

Device	Output-mirror transmission	Loaded Q	$d_2 = 2D_2/\kappa$	Confirmed pump-power range
A (primary)	0.12%	7.9×10^6	0.69	250 to 400 mW
B	0.25%	6.4×10^6	0.52	330 to 400 mW

At a common pump power of 350 mW, Device A shows a wide, stable single-soliton plateau on the thermally coupled branch (Figure 10) and a multi-soliton plateau on the idealized branch; Device B behaves comparably at a similar pump power with a somewhat larger output transmission, and is

retained as an alternative should the drop-port power of Device A prove too small to characterize directly (its intracavity field map is given in Supplementary Figure S4). A cavity with a still smaller output transmission (targeting a d_2 of 0.39 rather than 0.52) was found, and independently confirmed by seeding the simulation with an already-converged soliton state, to sit below the threshold for soliton existence at that lower d_2 ; a bound on this threshold, from the devices examined here, therefore falls between $d_2 = 0.39$ (soliton does not persist) and $d_2 = 0.52$ (soliton persists). Devices A and B, together with their gap and mirror-length tolerance variants, are laid out on the mask as Group E (Supplementary Table 5).

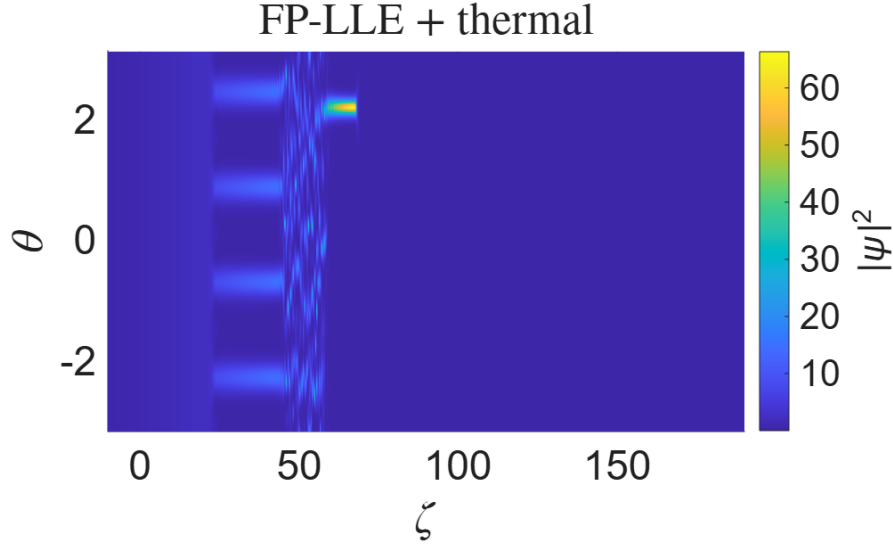


Figure 10: Intracavity field intensity of Device A (primary asymmetric design) across the detuning scan at $P_{\text{in}} = 350$ mW, thermally coupled branch. A single stable soliton peak forms after the modulation-instability region.

A direct time-domain sech-squared characterization, carried out the same way as for the reference device (Section 2.3), confirms this is a single-soliton state: the fit gives a 1182 fs pulse (fit quality 0.998) and a comb spanning 45 nm (Figure 12), somewhat narrower than the reference device's 59 nm, consistent with Device A's lower D_2 (8.45 MHz against 9.07 MHz).

2.5 Fabrication layout

2.5.1 Physical construction of a device

Each device (Figure 13) consists of an inverse-tapered edge coupler at each chip facet (narrowing to a $0.25\ \mu\text{m}$ tip for fiber coupling), a straight bus waveguide running from the edge coupler toward the cavity, the chirped Bragg mirror (a stepwise sequence of etched teeth, apodized at both ends), a short adiabatic taper joining the mirror's narrow apodized end to the wider spacer waveguide, the

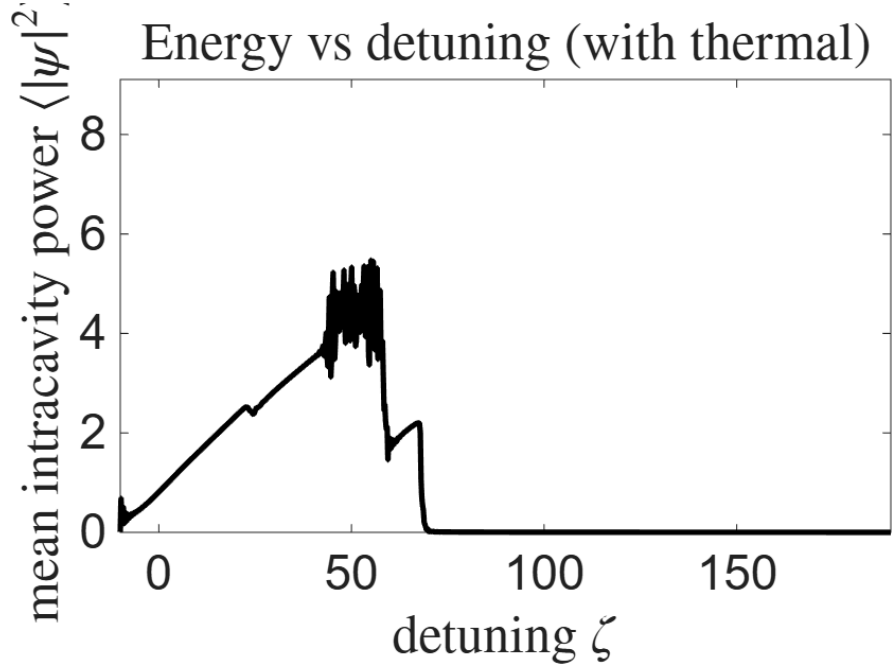


Figure 11: Mean intracavity intensity versus detuning for Device A at $P_{\text{in}} = 350$ mW (thermally coupled branch), showing the soliton plateau following the modulation-instability region.

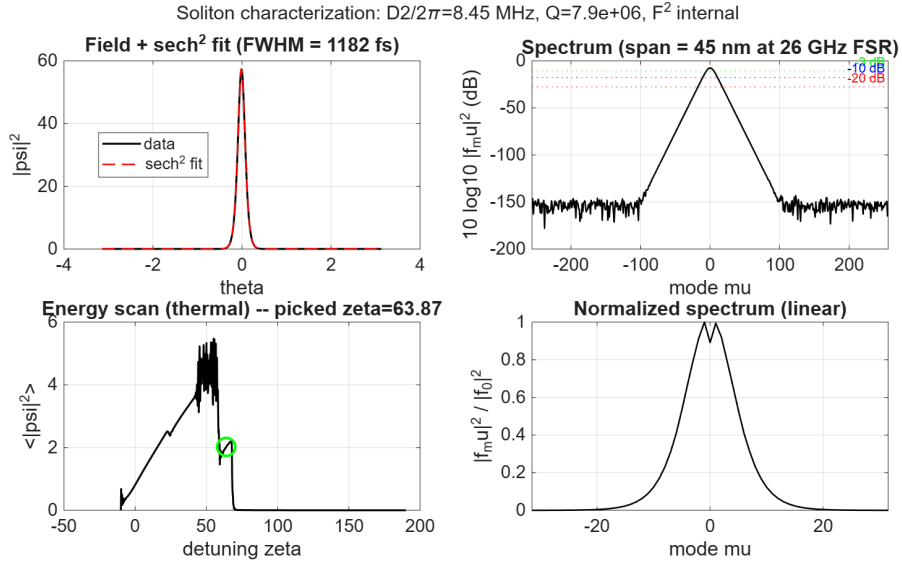


Figure 12: Time-domain sech-squared characterization of Device A (primary asymmetric design) at $P_{\text{in}} = 350$ mW, thermally coupled branch ($\zeta = 63.9$): intracavity field with the fitted sech-squared envelope (top left), comb spectrum with the $-3/-10/-20$ dB bandwidths marked (top right), the energy-versus-detuning scan with the characterized point marked (bottom left), and the normalized linear spectrum (bottom right).

uniform spacer, and a second taper and mirror pair (reversed chirp direction) closing the cavity, with a matching bus and edge coupler on the output side. Devices intended to be evanescently coupled to a bus (Groups B/C/D, E, and F, Subsection 2.5.2) additionally route that bus waveguide past the cavity at a fixed gap over the length of the spacer, using a pair of Euler S-bends to bring the bus from the edge-coupler line to the coupling gap and back; the direct-coupled devices of Group A omit the bus and gap entirely, exposing the cavity’s own two ports straight to the edge couplers, for measuring intrinsic device performance without a bus in the light path. The asymmetric devices of Group E and the devices supplied by Xinru (Group F) further add a third port on the transmissive (drop) mirror. The one-dimensional cavity model was translated into this layout using inverse-tapered edge couplers for fiber coupling, the taper geometry verified separately by simulation (Section 2.3), and a grating profile generator shared directly with the cavity solver.

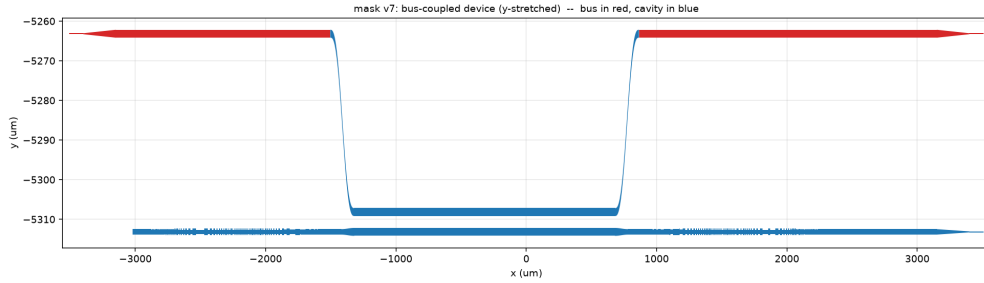


Figure 13: Bus-coupled device geometry as drawn on the current mask (bus waveguide in red, cavity in blue; vertical axis stretched for visibility), showing the Euler S-bend that brings the bus to the coupling gap alongside the cavity and back out to the edge coupler. All bus-coupled devices on the mask (Supplementary Tables 4-6) share this construction and differ only in cavity geometry and gap.

2.5.2 The current mask

The current mask integrates 72 devices on a single 7 mm by 12 mm chip, organized in four groups (parameters listed in Supplementary Tables 3 to 6): 14 direct-coupled (busless) symmetric cavities spanning the catalog of litho-safe geometries identified in the design search of Section 2.3, including the four in Table 1; 12 bus-coupled symmetric devices, built from three of those same catalog geometries (the reference device and Devices 1 and 2 of Table 1) and repeated across a sweep of bus-coupling gaps to bracket the critical-coupling point experimentally; 16 devices covering the two asymmetric drop-port designs of Table 2, each swept across bus gap and, for a subset, across mirror-length tolerance; and 30 devices supplied by Xinru, covering six base geometries each repeated across five bus gaps.

One item remains open at the mask level: the deep-ultraviolet (DUV) lithography process used to print these gratings has a linewidth bias that had not yet been recalibrated for the current process run. Every device in Table 1 was chosen with enough margin to clear the stepper’s resolution floor under a conservative estimate of this bias, and the mask is drawn at the target dimensions pending

mask v7 full chip overview (7 x 12 mm, 72 devices: 14A + 12B/C/D + 16E + 30F)
C:\Users\shimshra\Dropbox\shiwani_folder\fp_solitons\KLab_GDS_Lib-master\LibGDSpy\Designs\2026_06_SiN_CBG_FP_Soliton\cbg_fp_soliton_multichip_v7.gds

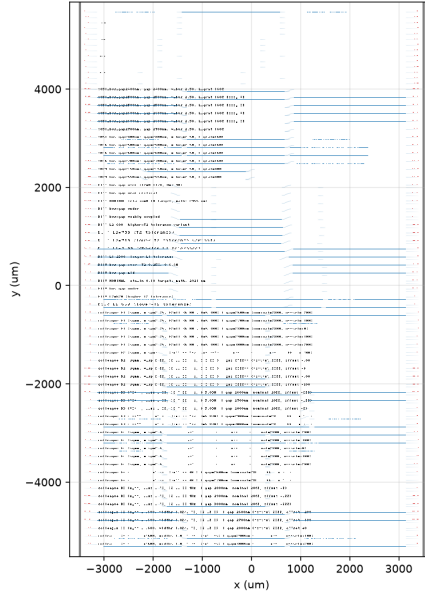


Figure 14: Full fabrication mask (7 mm by 12 mm, 72 devices): 14 direct-coupled symmetric cavities, 12 bus-coupled symmetric cavities, 16 asymmetric drop-port devices, and 30 devices supplied by Xinru, each row swept across a range of bus-coupling gaps.

that recalibration.

2.6 Discussion

Table 3 places the design reported here alongside the two studies discussed in Subsection 2.3.1 (Introduction) on the dispersion axis both identify as the central trade-off between comb-line power and comb bandwidth.

Table 3: Comparison with reported Fabry-Perot DKS microresonators.

Quantity	Wildi et al. [1]	Cheng et al. [2]	This work (reference device)
Mirror type	Uniform photonic-crystal reflector	Chirped, Blackman-apodized CBG	Chirped CBG, tanh-apodized
SiN core thickness	800 nm	777 nm	200 nm
Loaded / intrinsic Q	$\approx 4.0 \times 10^6$ (intrinsic)	≈ 4 to 5×10^6	1.03×10^7 (loaded)
$D_2/2\pi$	210 kHz	90 MHz	9.07 MHz
FSR	≈ 18.6 GHz	≈ 93 GHz	25.6 GHz
On-chip pump power	≈ 150 mW	≈ 410 mW	223 mW

The soliton pulse duration and comb span for the reference symmetric device and the primary asymmetric device have been confirmed by direct time-domain sech-squared characterization (Figure 6 and Figure 12); the remaining devices of Table 1 and Table 2 have so far only been confirmed through the width of the stable detuning plateau and the number of comb lines above the noise floor, which is consistent with, but less direct than, a sech-squared fit.

2.7 Conclusion

A chirped-Bragg-grating Fabry-Perot microresonator was designed in a thin silicon nitride waveguide platform. A coupled-mode cavity model and a standing-wave Lugiato-Lefever solver, including thermal feedback, were used to keep the second-order dispersion D_2 below 10 MHz, a regime that favors robust single-soliton formation, and four cavity geometries within that band were confirmed by direct simulation to support a stable soliton at loaded quality factors up to 1.03×10^7 , with fabrication tolerance to realistic lithographic width error verified for all four. An asymmetric extension of the same cavity, with one mirror made essentially fully reflective and the other given a small, deliberately constrained transmission, was designed to provide a dedicated soliton output port while keeping the loaded Q close to that of the validated symmetric design, and two such devices were confirmed by simulation to support a soliton at practical on-chip pump powers. All of the above have been laid out and verified on a single fabrication mask, the bus-to-cavity coupling simulation sweep of Section 2.3 was still running at the time of writing, and the next step is to fabricate and characterize these devices: measuring their transmission spectra and detuning-scan behavior, confirming the lithography process bias, and checking the bus-coupling behavior directly against the simulation reported here.

References

- [1] Wildi, T., Voumard, T., Brasch, V., Yilmaz, G. & Herr, T. Dissipative Kerr solitons in integrated Fabry-Perot microresonators. *Optica* **10**, 650-656 (2023).
- [2] Cheng, Y. et al. Communication-ready high-power soliton microcombs in highly dispersive Fabry-Perot microresonators. *Optica* **13**, 519-526 (2026).
- [3] Herr, T. et al. Temporal solitons in optical microresonators. *Nature Photonics* **8**, 145-152 (2014).
- [4] Kippenberg, T. J., Gaeta, A. L., Lipson, M. & Gorodetsky, M. L. Dissipative Kerr solitons in optical microresonators. *Science* **361**, eaan8083 (2018).
- [5] Lugiato, L. A. & Lefever, R. Spatial dissipative structures in passive optical systems. *Physical Review Letters* **58**, 2209-2211 (1987).

- [6] Cole, D. C., Gatti, A., Papp, S. B., Prati, F. & Lugiato, L. Theory of Kerr frequency combs in Fabry-Perot resonators. *Physical Review A* **98**, 013831 (2018).

Supplementary information

Supplementary Table 1. Fabrication tolerance of the four symmetric devices under a common ± 50 nm waveguide-width sweep (tooth-width contrast held fixed).

Device	Mirror length	D_2 peak-to-peak swing	Q peak-to-peak swing
1	800 μm	$< 3\%$	$\approx 28\%$
2	1600 μm	$< 3\%$	$\approx 3\%$
3 (reference)	1600 μm	$< 3\%$	$< 1\%$
4	1600 μm	$< 3\%$	$\approx 5\%$

For the reference device specifically, the same sweep on the original exploratory design point ($D_2/2\pi = 8.5$ MHz, $Q = 7.0 \times 10^6$, Subsection 2.3.3 discussion) gave a peak-to-peak D_2 variation of 0.55% and Q variation of 0.8% over the same ± 50 nm range, with mirror reflectivity and high-reflectivity bandwidth unchanged to within measurement precision.

Supplementary Table 2. Asymmetric drop-port design search, in order of decreasing output transmission.

Attempt	Output-mirror transmission	Loaded Q	$d_2 = 2D_2/\kappa$	Result
1	1.11%	2.99×10^6	0.21	Soliton does not persist (confirmed by seeded-state decay)
2	0.44%	5.11×10^6	0.39	Soliton does not persist (confirmed by seeded-state decay)
Device B	0.25%	6.41×10^6	0.52	Soliton confirmed, $P_{\text{in}} \geq 330$ mW
Device A	0.12%	7.90×10^6	0.69	Soliton confirmed, $P_{\text{in}} = 250$ to 400 mW

Supplementary Tables 3 to 6 list the parameters of all 72 devices on the current fabrication mask, reproduced directly from the mask build script’s device manifest and intended as the authoritative reference for this design (superseding any earlier partial listing). All devices use a duty cycle $D = 0.5$, tanh apodization with $W_{\text{flat}} = 0.6$, $a_{\text{roll}} = 0.08$, a 100 μm spacer-to-grating taper, and a litho pre-compensation factor of 1.0.

Supplementary Table 3. The 14 symmetric cavity geometries on the mask (Group A, direct-coupled, no bus), all with $L_{\text{cav}} = 2000$ μm , spacer width 2.5 μm , and chirp fraction 0.30.

The ten variants marked “no” in the last column were carried through the cavity-level design search of Section 2.3 (all meet the design’s D_2 , Q , and lithography constraints) but were not indi-

Variant	L_{grating} (μm)	Tooth widths w_1 / w_2 (μm)	Also bus-coupled?	Soliton-validated
v029	800	0.85 / 1.85	yes, 2 gaps	Device 1, Table 1
v033	800	1.00 / 2.50	no	no
v037	1200	0.85 / 1.60	no	no
v038	1200	0.85 / 1.85	no	no
v041	1200	1.00 / 2.00	no	no
v042	1200	1.00 / 2.50	no	no
v045	1200	1.15 / 2.65	no	no
v046	1600	0.85 / 1.60	yes, 3 gaps	Device 2, Table 1
v047	1600	0.85 / 1.85	no	no
v049	1600	1.00 / 1.75	no	no
v050	1600	1.00 / 2.00	no	no
v051	1600	1.00 / 2.50	yes, 7 gaps	Device 3 (reference), Table 1
v053	1600	1.15 / 2.15	no	no
v054	1600	1.15 / 2.65	no	Device 4, Table 1 (seeded scan)

vidually taken through soliton-dynamics validation; they are included on the mask as additional experimental data points bracketing the four validated designs.

Supplementary Table 4. Bus-coupling gap allocation for the three symmetric variants carried onto the bus-coupled mask (Group B/C/D, 12 devices), cavity geometry as in Supplementary Table 3.

Variant	Bus-coupling gaps (μm)	Number of devices
v051	2.40, 2.50, 2.50, 2.60, 2.60, 2.70, 2.80	7
v046	2.50, 2.60, 2.70	3
v029	2.45, 2.55	2

Supplementary Table 5. Group E: the two asymmetric drop-port designs and their gap and mirror-length tolerance sweeps (16 devices). Both share tooth widths 1.00/2.50 μm and spacer width 2.5 μm , total footprint ($L_1 + L_{\text{cav}} + L_2$) held at 5000 μm for all Device A variants by trading length between L_{cav} and whichever of L_1, L_2 is being varied.

Design	Nominal $L_1 / L_{\text{cav}} / L_2$ (μm)	Swept quantity	Values	Count
A (primary)	2200 / 2080 / 720	bus gap (μm)	2.60, 2.80, 3.00, 3.20, 3.40	5
A (primary)	as above	L_2 (μm), L_{cav} compensated	680, 700, 740	3
A (primary)	as above	L_1 (μm), L_{cav} compensated	2100, 2300	2
B (backup)	2200 / 2160 / 640	bus gap (μm)	2.40, 2.70, 2.90, 3.20	4
B (backup)	as above	L_2 (μm), L_{cav} compensated	620, 660	2

Supplementary Table 6. Group F: six designs (D1 to D6) supplied by Xinru, each repeated at bus gaps of 2.60, 2.70, 2.80, 2.90, and 3.00 μm (5 devices each, 30 total). All use tooth widths 1.15/2.65 μm except D6 (1.00/1.75 μm), and chirp fraction 0.30 on both mirrors.

Design	$L_1 / L_{\text{cav}} / L_2$ (μm)	L_{grating} (μm)	Spacer width (μm)	Symmetric?
D1	1200 / 2000 / 1200	1200	2.25	yes
D2	1200 / 2000 / 1200	1200	2.00	yes
D3	1600 / 2500 / 1200	1400	1.80	no (asymmetric)
D4	1200 / 2000 / 1200	1200	2.50	yes
D5	1200 / 2000 / 1200	1200	1.75	yes
D6	1600 / 2000 / 1600	1600	2.00	yes

D1, D2, D4, and D5 differ only in spacer width, mapping out its effect on this design point; D3 is the only asymmetric design in Group F, with independently chosen mirror lengths analogous to the two Group E designs of Table 2 but supplied and validated independently of this work.

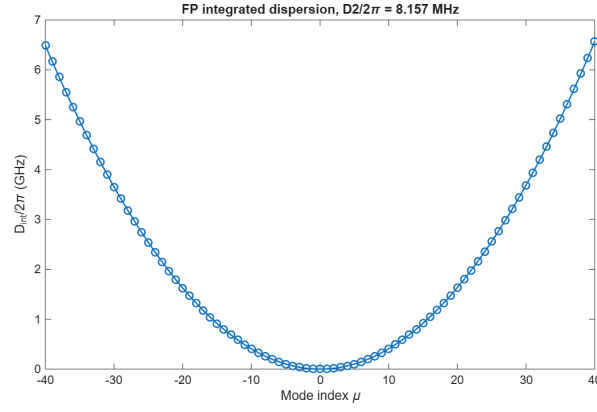


Figure 15: Supplementary Figure S1. Integrated dispersion of Device 1 (800 μm mirrors, $D_2/2\pi = 8.16$ MHz).

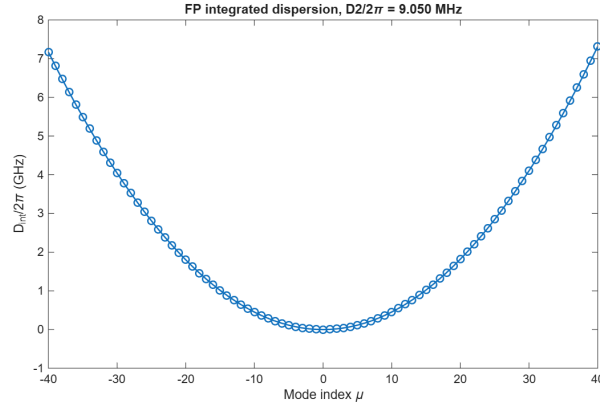


Figure 16: Supplementary Figure S2. Integrated dispersion of Device 2 (1600 μm mirrors, narrow tooth contrast, $D_2/2\pi = 9.05$ MHz).

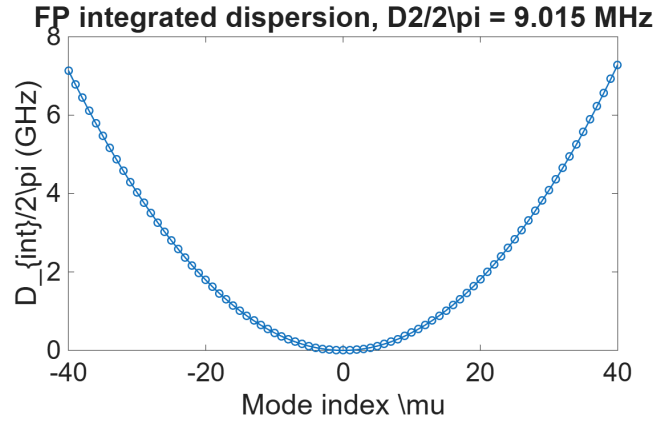


Figure 17: Supplementary Figure S3. Integrated dispersion of Device 4 (1600 μm mirrors, wide tooth contrast, $D_2/2\pi = 9.02$ MHz).

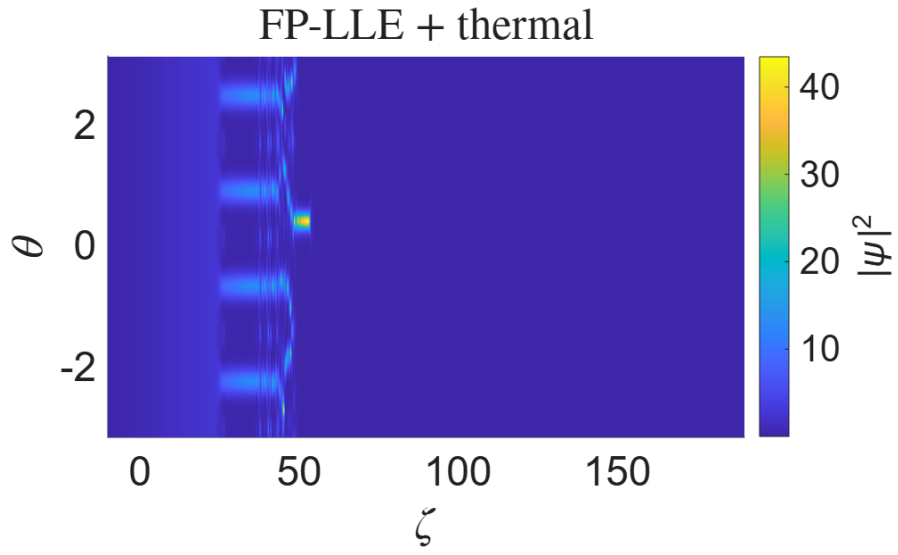


Figure 18: Supplementary Figure S4. Intracavity field intensity of Device B (backup asymmetric design) across the detuning scan at $P_{\text{in}} = 350$ mW, thermally coupled branch.

Chapter 3

Roughness-Limited Scattering Loss in Si_3N_4 Waveguides: Model Implementation and Validation Against a High-Confinement Photonic Platform

Abstract

We implement a group-usable model of surface-roughness scattering loss in Si_3N_4 optical waveguides, based on the equivalent-current formalism of Payne and Lacey extended to two-dimensional, high-index-contrast geometries by numerical evaluation of the mode-interface overlap. The implementation provides both a forward model (roughness statistics and mode overlap to loss) and two complementary inverse routes (measured loss to mode overlap, and measured loss to roughness), enabling any fabricated sample to be benchmarked against the model from geometry, a finite-element mode solution, and a measured intrinsic linewidth. We validate the model against the measured intrinsic loss of a 100 GHz-free-spectral-range (FSR) Si_3N_4 microring platform reported by Ji et al. [1], reproducing the reported top-and-bottom surface contribution to within the stated upper bound. Comparing the sidewall contribution obtained from a finite-element mode solution against the value required to reproduce the measured width-dependent linewidth, we find a persistent discrepancy of a factor 2.5 to 3.9, which we show is not attributable to mesh resolution, the ring bend radius, the roughness power spectral density convention, or the sidewall field decomposition. The residual discrepancy implies an effective sidewall roughness of 1.2 to 1.7 nm rather than the 0.75 nm value obtained by the original authors, a range consistent with their own scanning-electron-microscopy resolution limit. We further extend the model to 1 μm operation, obtaining the waveguide group index and mode overlap from a finite-element re-evaluation and

finding the resonator bend radius requires no meaningful change; the absorption contribution at this wavelength remains to be measured.

3.1 Introduction

As absorption losses in silicon nitride (Si_3N_4) photonic integrated circuits (PICs) have been reduced through improved deposition and annealing protocols, radiation loss from scattering by residual sidewall and interface roughness has become the dominant remaining loss channel in tightly confined, high-index-contrast waveguides [1]. Predicting this loss from geometry and roughness statistics, or conversely inferring an effective roughness from a measured loss, is therefore of direct practical value for benchmarking fabricated devices and for informing waveguide design.

The original theoretical treatment of waveguide scattering loss, due to Marcuse [2], modeled the loss as coupling between the fundamental guided mode and the continuum of radiation modes. The resulting expression requires a complete near- and far-field description, which obscures the dependence of the loss on the waveguide's physical parameters. Payne and Lacey [3] reformulated the problem using the method of equivalent currents, in which only the far field responsible for radiation loss is retained, and obtained a closed-form expression for the scattering loss coefficient of a symmetric, single-mode planar (one-dimensional) waveguide with either an exponential or a Gaussian roughness autocorrelation function (ACF). Their result separates the loss into a purely geometric factor and a spectral factor that depends on the roughness correlation length relative to the beat length between the guided mode and the radiation continuum, and shows that the loss has a rigorous upper bound independent of the detailed roughness statistics.

For real two-dimensional, high-index-contrast waveguides, the modal field at each rough interface is not available in closed form and must instead be obtained from a full-vectorial mode solution. Ji et al. [1] applied this extension to a Si_3N_4 microring platform (720 nm core height, 1.7 to 2.6 μm width, 100 GHz FSR), evaluating the equivalent-current formula with a finite-element mode solution of the bent waveguide and reporting a resulting sidewall roughness of 0.75 nm consistent with their measured intrinsic linewidth as a function of width.

This report documents an independent implementation of this modeling framework: the closed-form one-dimensional model, its two-dimensional finite-element-based extension, and two inverse routes that recover either the mode overlap or the roughness from a measured loss. We validate the implementation against the measured dataset of Ji et al. [1], investigate a discrepancy between the finite-element-predicted and the measurement-required sidewall mode overlap, and report on ongoing work extending the model to 1 μm operation.

3.2 Theory

3.2.1 One-dimensional equivalent-current model

For a symmetric planar waveguide of core index n_1 , cladding index n_2 , half-thickness d , and propagation constant β , Payne and Lacey [3] give the power attenuation coefficient due to roughness scattering at both walls as

$$\alpha = \varphi^2(d) (n_1^2 - n_2^2)^2 \frac{k_0^3}{4\pi n_1} \int_0^\pi \tilde{R}(\beta - n_2 k_0 \cos\theta) d\theta, \quad (1)$$

where k_0 is the free-space wavenumber, $\varphi(d)$ is the normalized modal field evaluated at the core-cladding interface, and $\tilde{R}(\Omega)$ is the roughness power spectral density (PSD), the Fourier transform of the roughness autocorrelation function $R(u)$. The roughness is characterized by its root-mean-square (RMS) height σ and correlation length L_c , with $\sigma^2 = R(0)$.

For an exponential ACF, $R(u) = \sigma^2 \exp(-|u|/L_c)$, the PSD integral in (1) has a closed form. Writing $M = 2\beta L_c$ and $N = 1 - L_c^2(\beta^2 - n_2^2 k_0^2)$,

$$S \equiv \int_0^\pi \tilde{R}(\beta - n_2 k_0 \cos\theta) d\theta = \sqrt{2} \sigma^2 L_c \pi \frac{[\sqrt{M^2 + N^2} + N]^{1/2}}{\sqrt{M^2 + N^2}}. \quad (2)$$

Introducing the normalized waveguide parameters $V = k_0 d \sqrt{n_1^2 - n_2^2}$, U , W (the usual slab-waveguide normalized transverse wavenumbers), the normalized correlation length $x = W L_c / d$, and the guiding-strength parameter $\gamma = n_2 V / (n_1 W \sqrt{\Delta})$ with $\Delta = (n_1^2 - n_2^2) / 2n_1^2$, (1) can be written in the separable form

$$\alpha = \frac{\sigma^2}{\sqrt{2} k_0 d^4 n_1} g(V) f_e(x, \gamma), \quad (3)$$

where $g(V) = U^2 V^2 / (1 + W)$ depends only on waveguide geometry and $f_e(x, \gamma)$ depends only on the roughness statistics through x and γ . For a fixed γ , $f_e(x, \gamma)$ has a single maximum in x : the loss is largest when the roughness correlation length is comparable to the beat length between the guided mode and the radiation continuum, and falls off on either side of this resonance. Most practical waveguides operate with $\gamma \gg 1$ (weak guidance) and x well below the resonance ($x \lesssim 5$), so that the loss is a slowly varying, monotonic function of L_c in the regime relevant to fabricated devices; this has direct consequences for the identifiability of σ and L_c separately from a single measured loss curve, discussed in Subsection 3.4.2.

3.2.2 Extension to two-dimensional waveguides

For a two-dimensional waveguide cross-section, the analytic slab field $\varphi^2(d)$ is replaced by a normalized line-to-area overlap integral evaluated numerically from a full-vectorial mode solution,

$$\Gamma = \frac{\int_{\text{interface}} I(x, y) d\ell}{\iint_{\text{area}} I(x, y) dA}, \quad (4)$$

where I is the modal intensity. The loss contribution of a given rough interface, following the same equivalent-current derivation, is

$$\alpha_i = \Gamma_i (n_{\text{eff}}^2 - n_{\text{clad},i}^2)^2 \frac{k_0^3}{4\pi n_{\text{eff}}} S_i, \quad (5)$$

with S_i given by (2) evaluated at the waveguide's effective index n_{eff} in place of the slab index, and n_{eff} replacing n_1 in the prefactor. Contributions from statistically independent, nominally identical interfaces (e.g. two sidewalls) add linearly, each multiplied by its multiplicity. Converting the total attenuation coefficient $\alpha = \sum_i \alpha_i$ to a resonator linewidth uses the group index n_g , not the phase index: $\kappa/2\pi = (c_0/n_g) \alpha/2\pi$, where c_0 is the vacuum speed of light.

A finite waveguide height reduces sidewall scattering relative to the infinite-height slab assumed in (5). Barwicz and Haus [4] derive a correction factor $C_{3D} \leq 1$ for vertical sidewalls of finite height, obtained from the same transverse-mode dispersion relation applied to the vertical (height) direction; we include this correction as an option in the implementation, discussed further in Subsection 3.4.2.

3.3 Methods

Mode overlap. The effective index n_{eff} , group index n_g , and mode overlap Γ at each interface (top, bottom, and the two sidewalls) were obtained from a full-vectorial finite-element mode solution of the waveguide's actual bent geometry, using a two-dimensional axisymmetric model revolved about the ring axis at the resonator's physical bend radius. Because the mode shifts outward in a bend, a straight-waveguide mode solution would systematically misstate the sidewall overlap, particularly the asymmetry between the inner and outer sidewalls; the axisymmetric bent-mode solution captures this directly. Cladding dispersion was taken from the standard Sellmeier fit for thermally grown, fused-silica-type SiO_2 [5]. Core dispersion was taken from a three-term Sellmeier fit for LPCVD Si_3N_4 already in use for this platform's finite-element model prior to this work. The group index was obtained by a central difference of the effective index across a narrow wavelength window centered on the wavelength of interest. Mesh convergence, including refinement at the four corners of the rectangular core cross-section, was verified by comparing successive mesh refinements (Subsection 3.4.2).

Forward model. Given n_{eff} , n_g , the cladding index, and, for each interface, Γ , σ , L_c , and the ACF (5) gives the loss contribution of that interface; the exponential ACF (2) was used throughout, consistent with atomic-force-microscopy (AFM) characterization of LPCVD Si_3N_4 and thermal SiO_2 surfaces reported for this platform [1].

Inverse routes. Two complementary inversions were implemented. First, because (5) is linear in Γ , a measured total loss with σ and L_c assumed (or independently known) can be inverted exactly for Γ once all other loss contributions are subtracted:

$$\Gamma = \frac{\alpha_{\text{meas}} - \alpha_{\text{known}}}{(n_{\text{eff}}^2 - n_{\text{clad}}^2)^2 k_0^3 / (4\pi n_{\text{eff}}) S}. \quad (6)$$

Second, because S , and hence α , is exactly quadratic in σ at fixed L_c (2), a measured loss with Γ and L_c known can be inverted exactly for σ by evaluating the forward model once at a unit probe value of σ to obtain the proportionality constant. Both inversions were verified by round-trip reconstruction of synthetic data to floating-point precision.

Validation dataset. We validate against the reported intrinsic linewidth $\kappa_0/2\pi$ at 1550 nm for four waveguide widths (1.7, 2.0, 2.3, and 2.6 μm) of the 100 GHz Si_3N_4 microring platform of Ji et al. [1], together with their independently measured absorption loss ($\kappa_{\text{abs}}/2\pi = 4$ MHz, obtained by a Kerr-calibrated thermal technique and reported as approximately width-independent) and AFM-characterized top and bottom surface roughness ($\sigma = 0.25$ nm, $L_c = 13$ nm).

3.4 Results

3.4.1 Top and bottom surface scattering

Using the AFM-measured top and bottom roughness together with the finite-element mode overlap, the top-plus-bottom contribution to the intrinsic linewidth is 0.44 to 0.48 MHz across the measured width range, consistent with the platform's own reported bound ($\kappa_{ts} + \kappa_{bs} < 0.5$ MHz [1]).

3.4.2 Sidewall scattering: finite-element prediction versus the value required by measurement

Two independent estimates of the sidewall mode overlap Γ_{side} were compared. The first was obtained directly from the finite-element mode solution described in the Methods section. The second was obtained by inverting (6) using the measured $\kappa_0(w)$ of Ref. [1], after subtracting the absorption and top/bottom contributions, with $\sigma = 0.75$ nm and $L_c = 75$ nm (the mid-range of the 50 to 100 nm bracket reported in Ref. [1]).

Table 1: Sidewall mode overlap: finite-element versus back-calculated (assuming $\sigma = 0.75$ nm, $L_c = 75$ nm).

Width (μm)	$\Gamma_{\text{side, FEM}}$ (m^{-1})	$\Gamma_{\text{side, back-calculated}}$ (m^{-1})	Ratio
1.7	9.72×10^4	3.76×10^5	0.259
2.0	6.37×10^4	2.07×10^5	0.308
2.3	4.41×10^4	1.29×10^5	0.342
2.6	3.19×10^4	7.98×10^4	0.400

The finite-element value is 2.5 to 3.9 times smaller than the value required to reproduce the measured linewidth, and the ratio is not constant with width. Figure 1 shows the resulting total intrinsic linewidth predicted from the finite-element Γ_{side} , with and without the Barwicz-Haus finite-height correction ($C_{3D} = 0.794$ at this core height and wavelength), against the measured values; both finite-element predictions fall well below the measurement at every width, and applying the finite-height correction, which is physically well motivated, increases rather than decreases the discrepancy.

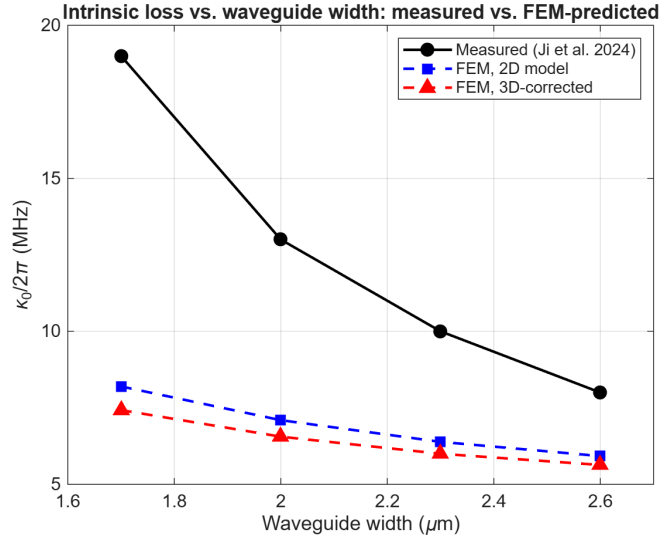


Figure 1: Measured intrinsic linewidth (Ref. [1]) compared to the finite-element prediction with $\sigma = 0.75$ nm, with and without the Barwicz-Haus finite-height correction.

3.4.3 Investigation of the discrepancy

We tested four candidate explanations for the discrepancy in Table 1 and found each insufficient to account for it.

Uncertainty in the correlation length L_c (50 to 100 nm) changes the back-calculated Γ_{side} by

Table 2: Candidate explanations for the finite-element-versus-measurement sidewall discrepancy.

Candidate cause	Test	Result
Mesh resolution at the core corners	Compared successive mesh refinements down to 1 nm minimum element size at the four corners of the rectangular core	Changes Γ by 0.6%; ruled out
Ring bend radius	Compared the platform's 231 μm (100 GHz) bend radius against a 454 μm (50 GHz) radius	Changes Γ by less than 1%; ruled out
Roughness PSD convention (exponential vs. Gaussian ACF)	Cross-checked the closed-form exponential PSD (2) against the model definition	Confirmed correctly implemented; ruled out
Sidewall field decomposition (which field component, and which side of the interface, defines Γ)	Compared the normal and tangential field components on the core side, the cladding side, and their average, using the finite-element field decomposition at each sidewall	The cladding-side field alone gives a Γ up to 1.34 times larger than the core/cladding average used above, accounting for at most a third of the discrepancy

only about $\pm 8\%$, far smaller than the observed factor of 2.5 to 3.9, and does not by itself explain the discrepancy either.

Given that the discrepancy is not attributable to numerical convergence, geometry, or convention, we instead ask what sidewall roughness a purely finite-element-based model would require to match the measurement directly, without back-calculating through the inversion. Figure 2 shows the resulting σ , with and without the finite-height correction, compared to the sidewall roughness upper bound obtained by Ref. [1] from scanning electron microscopy (SEM), 1 to 1.5 nm, itself limited by the SEM's 1.33 nm per-pixel resolution.

The required roughness, 1.18 to 1.66 nm depending on width and on whether the finite-height correction is applied, falls within or just above this SEM-limited upper bound. This is consistent with an actual sidewall roughness larger than the 0.75 nm value obtained by Ref. [1], whose own reported SEM measurement provides only an upper bound rather than a direct measurement of σ .

3.4.4 Extension toward 1 μm operation (in progress)

Work is in progress to extend the validated model to a device operating near 1 μm . A 1 μm design requires re-deriving the bend radius, since the free spectral range depends on wavelength through the group index:

$$\text{FSR} = \frac{c_0}{n_g L}, \quad L = 2\pi R, \quad (7)$$

with L the round-trip path length. Re-evaluating the core and cladding Sellmeier dispersion at 1 μm gives $n_{\text{core}} = 2.0009$ and $n_{\text{clad}} = 1.4504$ (compared to 1.9867 and 1.4440 at 1550 nm), and the

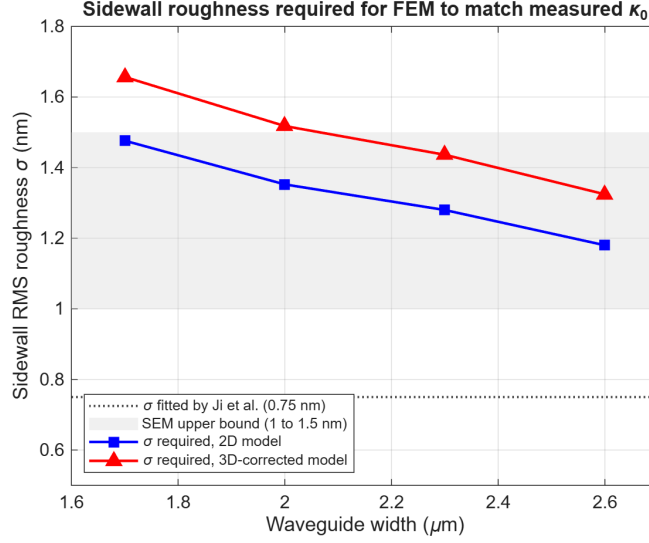


Figure 2: Sidewall RMS roughness required for the finite-element model to reproduce the measured $\kappa_0(w)$ directly, without the linear Γ -inversion, compared to the fitted value of Ref. 1 and its own reported SEM resolution limit.

vertical mode confinement parameter $V_{\text{vert}} = k_0(h/2)\sqrt{n_{\text{core}}^2 - n_{\text{clad}}^2}$ increases from 1.99 to 3.12 at the platform's 720 nm core height, indicating a more strongly multi-moded vertical confinement at the shorter wavelength.

A finite-element mode solve at 1 μm was completed using the platform's existing 231 μm bend radius and a corrected azimuthal mode number for the new wavelength. Re-deriving the bend radius from (7) with the resulting group index gave $R = 231.406 \mu\text{m}$, a 0.18% change from the 1550 nm value. The finite-element model was re-run a second time at this corrected radius (with the azimuthal mode number re-derived accordingly) to confirm self-consistency; the resulting group index changed by no more than 0.0003% relative to the first pass, confirming the radius-insensitivity already established in Subsection 3.4.2 (Table 2) and validating $R = 231.406 \mu\text{m}$ as the self-consistent bend radius at this wavelength. The results below are from this corrected-radius run.

Table 3 gives the resulting scattering-only intrinsic linewidth (top, bottom, and sidewall contributions) at the four widths used throughout this report, with and without the Barwicz-Haus finite-height correction ($C_{3D} = 0.8895$ at 1 μm , versus 0.794 at 1550 nm, consistent with the larger V_{vert} at the shorter wavelength). Figure 3 shows the same result together with the $L_c = 50$ to 100 nm sensitivity band.

The absorption contribution to the intrinsic linewidth at 1 μm has not been measured and is required to complete the loss budget at this wavelength.

Table 3: Predicted scattering-only intrinsic linewidth at 1 μm , $R = 231.406 \mu\text{m}$ (no measured $\kappa_0(w)$ exists yet at this wavelength for comparison).

Width (μm)	$\kappa_{\text{scatter}}/2\pi$, 2D (MHz)	$\kappa_{\text{scatter}}/2\pi$, 3D-corrected (MHz)
1.7	10.020	9.096
2.0	7.241	6.630
2.3	5.663	5.229
2.6	4.738	4.408

Intrinsic scattering loss vs width at 1 μm -- FEM 2D vs 3D-corrected (prediction, no measurement exists yet)

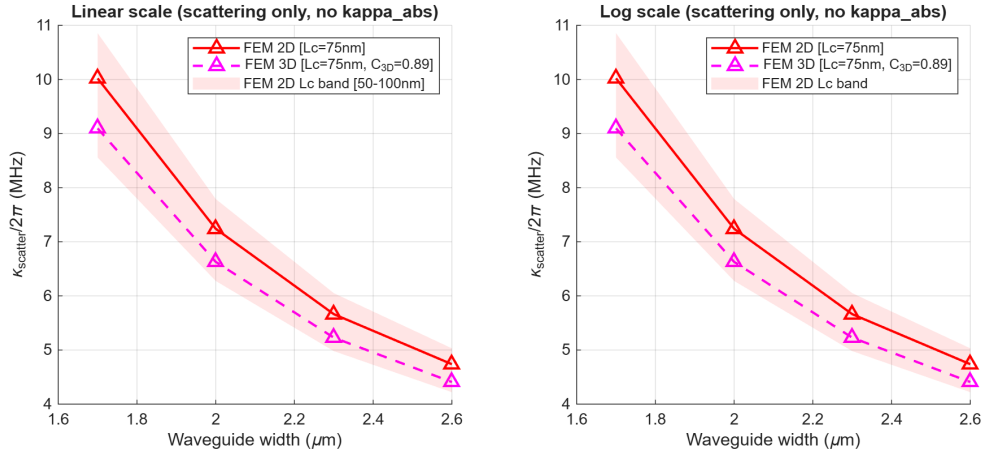


Figure 3: Predicted scattering-only intrinsic linewidth at 1 μm as a function of waveguide width, from the finite-element model (2D and 3D-corrected), with the $L_c = 50$ to 100 nm sensitivity band shaded.

3.5 Discussion

The top-and-bottom-surface contribution to the intrinsic linewidth is reproduced by the model to within the platform's own reported bound using an unmodified finite-element mode overlap, confirming that the master formula (5) and its numerical implementation are internally consistent for at least one class of interface. For the sidewall contribution, however, the finite-element mode overlap and the value required by measurement disagree by a factor that grows systematically with width (2.5 at 1.7 μm to 3.9 at 2.6 μm), a pattern inconsistent with a fixed multiplicative error and instead suggestive of a genuine, width-dependent physical effect: as the waveguide becomes wider and more strongly confined, the sidewall field intensity computed by the finite-element solver becomes smaller relative to what a fixed roughness would need to produce the measured loss.

The most direct interpretation is that the actual sidewall RMS roughness on this platform exceeds the 0.75 nm value reported by Ref. [1]. That value was itself obtained as a fit that makes the authors' own finite-element-based model match their measured linewidth, not as a direct measurement;

their independent SEM characterization gives only an upper bound of 1 to 1.5 nm, limited by the microscope’s pixel resolution, which is consistent with the 1.18 to 1.66 nm range we find is required here. Applying the Barwicz-Haus finite-height correction, which is physically justified for this waveguide’s finite core height, moves the required roughness further from 0.75 nm rather than closer to it, which argues against attributing the discrepancy to the two-dimensional roughness model’s known overestimate for finite-height sidewalls [1,4]. We could not identify a numerical or definitional artifact (mesh convergence, bend radius, PSD convention, or field-component choice) that accounts for a large fraction of the gap; the sidewall field decomposition test accounts for at most a third of it. A conclusive resolution would require an independent, higher-resolution measurement of the sidewall roughness, for example transmission electron microscopy in cross-section.

Despite this open discrepancy in the absolute finite-element-based sidewall overlap, the two inverse routes described in the Methods section remain useful for benchmarking a newly fabricated sample: given a measured intrinsic linewidth as a function of width, the linear Γ -inversion (6) recovers a self-consistent effective mode overlap without requiring the absolute finite-element value to be correct, and the quadratic σ -inversion recovers an effective roughness given an independently trusted mode overlap. We note, however, that attempting to recover both σ and L_c simultaneously from a single width sweep is not reliable for this waveguide geometry: because these waveguides operate on the shallow, low- x side of the resonance in $f_e(x, \gamma)$ (see the theory section) rather than near its peak, a joint nonlinear fit of σ and L_c to the same four-point dataset is essentially unconstrained in L_c , converging to correlation lengths many orders of magnitude beyond any physically plausible value for a negligible improvement in fit quality. We therefore recommend fixing L_c from an independent source (AFM or SEM) and fitting only σ , reporting a bracket over the plausible range of L_c where an independent value is not available.

3.6 Conclusion

We implemented and validated a two-dimensional, finite-element-based extension of the Payne-Lacey equivalent-current model for roughness scattering loss in Si_3N_4 waveguides, together with two exact inverse routes for benchmarking fabricated samples from a measured intrinsic loss. The model reproduces the top-and-bottom surface contribution to a validated Si_3N_4 microring platform to within its reported bound. For the sidewall contribution, we find and characterize a persistent, width-dependent discrepancy of a factor 2.5 to 3.9 between the finite-element-predicted and measurement-required mode overlap, which we attribute most plausibly to an actual sidewall roughness larger than the previously reported 0.75 nm, consistent with that platform’s own SEM resolution limit, rather than to a numerical or definitional error in the model. The model was extended to 1 μm operation via a direct finite-element re-evaluation of the waveguide group index and mode overlap; the resonator bend radius requires no meaningful change from the 1550 nm design. The absorption contribution at 1 μm has not been measured and is needed to complete the

loss budget at this wavelength.

References

- [1] X. Ji, R. N. Wang, Y. Liu, J. Riemensberger, Z. Qiu, and T. J. Kippenberg, “Efficient mass manufacturing of high-density, ultra-low-loss Si_3N_4 photonic integrated circuits,” *Optica* **11**, 1397-1407 (2024).
- [2] D. Marcuse, “Radiation losses of the dominant mode in round dielectric waveguides,” *Bell Syst. Tech. J.* **48**, 3187 (1969).
- [3] F. P. Payne and J. P. R. Lacey, “A theoretical analysis of scattering loss from planar optical waveguides,” *Opt. Quantum Electron.* **26**, 977-986 (1994).
- [4] T. Barwicz and H. A. Haus, “Three-dimensional analysis of scattering losses due to sidewall roughness in microphotonic waveguides,” *J. Lightwave Technol.* **23**, 2719 (2005).
- [5] I. H. Malitson, “Interspecimen comparison of the refractive index of fused silica,” *J. Opt. Soc. Am.* **55**, 1205-1208 (1965).

Supplementary information

S1. Extended finite-element mode data. Effective index, group index, and per-interface mode overlap at 1550 nm, from the axisymmetric bent-waveguide finite-element solution used throughout the Results section (231 μm bend radius).

Width (μm)	n_{eff}	n_g	$\Gamma_{\text{top}} (\text{m}^{-1})$	$\Gamma_{\text{side, inner}} (\text{m}^{-1})$	$\Gamma_{\text{side, outer}} (\text{m}^{-1})$
1.7	1.8132	2.0973	4.567×10^5	8.899×10^4	1.055×10^5
2.0	1.8244	2.0896	4.604×10^5	5.539×10^4	7.208×10^4
2.3	1.8320	2.0842	4.624×10^5	3.567×10^4	5.252×10^4
2.6	1.8373	2.0803	4.637×10^5	2.347×10^4	4.041×10^4

The inner and outer sidewalls differ by 19% at 1.7 μm width, growing to 72% at 2.6 μm , reflecting the outward shift of the mode centroid in the bend; the values in Table 1 are the average of the two.

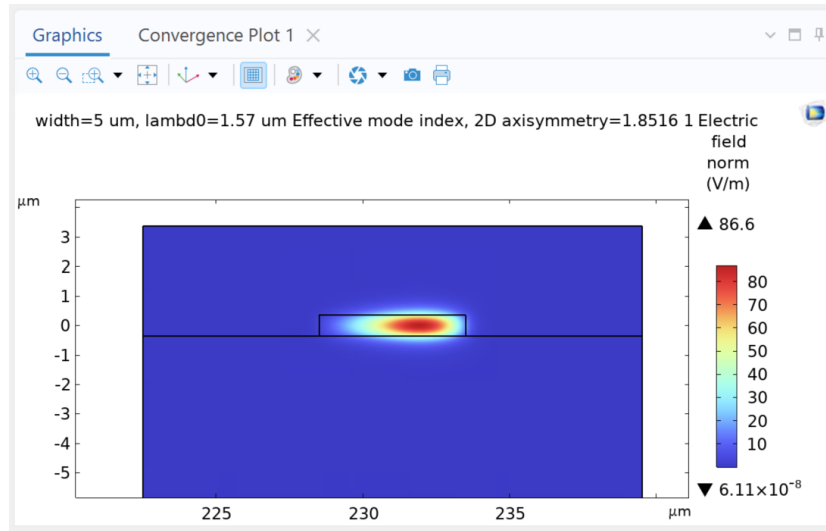


Figure 4: Supplementary Figure S1. Simulated fundamental mode profile (electric field norm) of the axisymmetric bent-waveguide finite-element model, for a 5 μm -wide waveguide at 1.57 μm , giving an effective index of 1.8516. Shown for illustration of the mode's confinement within the 720 nm core with an evanescent tail into the cladding, consistent with the low sidewall field intensity discussed in the Discussion.

Chapter 4

Technical Notes

Alongside the two design projects, three technical notes (TNs) were contributed to LPQM's internal technical-notes collection.

4.1 TN 128: PowerPoint Vectorized Search

TN 128, co-authored with Marco Golino, Shreya Shah, and Alessio Zicoschi, documents the PowerPoint semantic search tool: its architecture, author-detection logic, database design, and the Cloudflare Zero Trust setup used to reach it from outside the EPFL network.

Laboratory of Photonics and Quantum Measurements, EPFL
Prof. Dr. Tobias J. Kippenberg

Technical Note 128 **1**

PowerPoint Vectorized Search

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Laboratory of Photonics and Quantum Measurements (LPQM)

1 Overview

This note documents the design and implementation of a hybrid semantic search system for PowerPoint presentations. The system lets users search across an archive of .pptx files using natural language. Results are ranked by a combination of cosine similarity over dense sentence embeddings

4.2 TN 129: Introduction to the KLab GDS Library

TN 129 documents `KLab_GDS_Lib`, the group's Python toolkit for drawing photonic-integrated-circuit (PIC) masks. It covers environment setup and two worked examples, for anyone in the group laying out a mask for the first time.

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Technical Note 129

1

Introduction to the KLab GDS Library

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1 Introduction

`KLab_GDS_Lib` is a Python toolkit for designing photonic integrated circuit (PIC) masks. It wraps the open-source `gdsapy` [1] package with three higher-level abstractions:

4.3 TN 130: Scripting Lumerical simulations with LSF files

TN 130 documents the LSF (Lumerical Script File) scripting workflow as an alternative to driving Lumerical through its GUI, covering a worked FDTD example, converting a GUI-built project into a script, and pairing LSF scripting with AI coding assistants.

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Technical Note 130

1

Scripting Lumerical simulations with LSF files

Shiwani Mishra

Swiss Federal Institute of Technology (EPFL)

Laboratory of Photonics and Quantum Measurements (LPQM)

1 Introduction

Most people in the group are already comfortable driving Lumerical through its GUI. The scripting side is used less often, which is what this note is about: adding a text-based workflow alongside the GUI, not replacing it.

Chapter 5

Software tools

Three small software tools were built to support tasks that come up repeatedly in day-to-day lab work: finding a past presentation, finding a past technical note, and finding the files associated with a fabrication design.

5.1 PPTX semantic search

The lab accumulates presentations (research updates, group meetings, conference talks) across a shared network drive, and finding a specific slide from memory of what it said, rather than which file or folder it was saved in, was not previously possible. This tool provides a search box, reached from any browser at a fixed address, that lets a lab member type a natural-language query (e.g. “vortex dynamics” or “soliton crystal linewidth”) and get back the presentations and slides that match, ranked by a combination of semantic similarity and keyword matching. Each result can be opened to preview the matching slide or downloaded as the original `.pptx` file. No installation is required on the user’s side; a single always-on machine indexes the shared drive’s presentations and serves the search interface to everyone else. New presentations are picked up the next time the archive is re-indexed, which currently happens manually when new files are added.

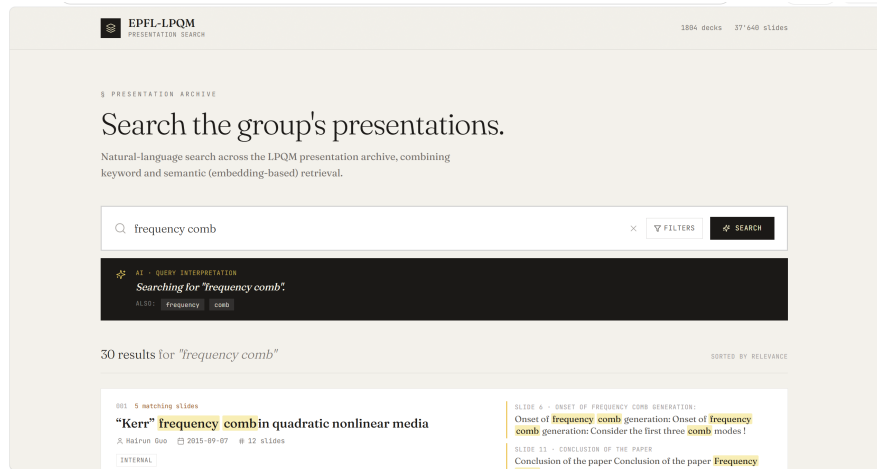


Figure 5: PPTX semantic search web interface. The query “frequency comb” returns 30 matching presentations, ranked by relevance, with an AI-generated query interpretation and the specific matching slides listed for each result.

5.2 LPQM technical-notes catalog

The lab’s technical-notes collection is version-controlled on GitLab and includes notes contributed by lab members across topics ranging from nanofabrication procedures to simulation workflows. A generated catalog page lets a lab member search this collection by title, author, or tag, browse notes by category, see a thumbnail preview of each note, and open its PDF directly, all from a single downloadable HTML page that works without any server. A documented template lets a new contributor add their own note and regenerate the catalog with a short, repeatable process.

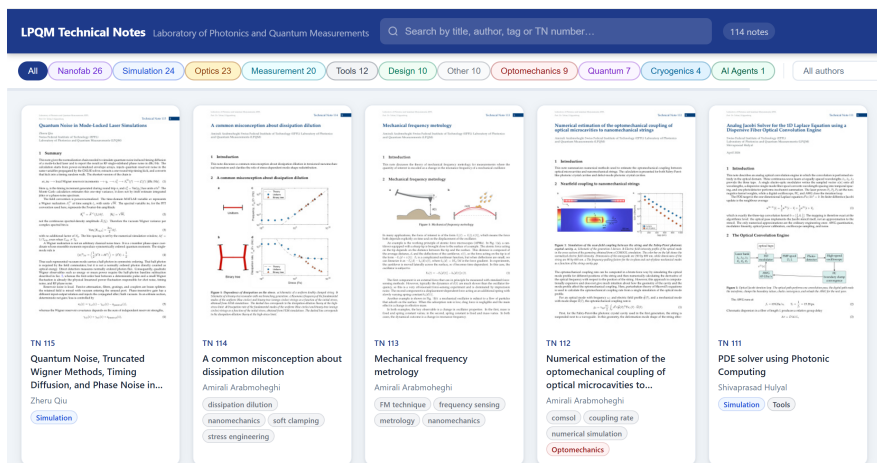


Figure 6: LPQM technical-notes catalog page: a searchable, tag-filterable list of notes, each shown with a thumbnail of its first page, title, author, and tags.

5.3 Fab log UI

Fabrication design files (mask layouts, GDS files, design documentation, and run cards) accumulate in per-design folders on the group’s shared drive under an internal naming convention. This tool provides a single browsable, filterable table view of every design folder: a lab member can filter by design number, project name, or point of contact, and see at a glance which of the standard file categories (design documentation, GDS layout, characterization data, run card) are present for each design, without navigating the shared drive’s folder structure by hand. It runs as a lightweight local page refreshed periodically from a scan of the shared drive, so it stays current as new designs are added.

LPQM Comb Designs

Front end to the design-file share.

229 designs. Generated 2026-06-18.

Filter by design number, name, or POC (e.g. D247, modulator, Xinru)...

Drive location: \\sb-nas1.rcp.epfl.ch\LPQM1\group\GROUP_SHARE\Comb\Samples

Network (mb) This Mac (/Volumes/LPQM1)

blue = direct file

green = folder (multiple files)

grey = not found

PPT documentation, GDS layout, lightning candle, runcard

DESIGN	PROJECT	POC	CREATED	FILES
D254	LXOD Transducer	Wil, Xiaoyi	Jun 2026	PPT GDS CANDLE RUNCARD FOLDER
D253	Dissipative engineering soliton	Yamau, Alisa	Jun 2026	PPT GDS CANDLE RUNCARD
D252	MLL Hybrid DualComb	Xuan, Xurong, Zheru	Jun 2026	PPT GDS CANDLE RUNCARD
D251	Multilane transmitter	Yichi	—	PPT GDS CANDLE RUNCARD
D250	AML second run	Yichi	Jun 2026	PPT GDS CANDLE RUNCARD
D249	MLL FreqSweeping	Xuan	Jun 2026	PPT GDS CANDLE RUNCARD
D248	MLL 1480WDM	Yamau, Zheru	Jun 2026	PPT GDS CANDLE RUNCARD
D247	Superconductive Long modulator	Xiaoyi, Zihan	Jun 2026	PPT GDS CANDLE RUNCARD
D246	EOsampling	Zheru, Junyin	Jun 2026	PPT GDS CANDLE RUNCARD FOLDER
D245	TR cavity EO rerun	Jiachen, Junyin	Jun 2026	PPT GDS CANDLE RUNCARD FOLDER

Figure 7: Fab log UI: a filterable table of fabrication designs, showing which file categories are present for each.

Chapter 6

Conclusion

This internship produced two validated photonic designs and a set of supporting contributions to LPQM's shared infrastructure. The chirped-Bragg-grating Fabry-Perot microresonator project (Chapter 2) identified four symmetric cavity geometries and two asymmetric drop-port geometries, all confirmed by direct soliton-dynamics simulation to support a stable dissipative Kerr soliton at practical on-chip pump powers, tolerant to realistic lithographic width error, and laid out on a single fabrication mask ready for the group's next fabrication run. The scattering-loss modeling project (Chapter 3) delivered a validated, reusable forward and inverse model for roughness-limited loss in Si_3N_4 waveguides, reproducing a published platform's top-and-bottom surface loss to within its own reported bound. On that basis, the model was extended to $1\text{ }\mu\text{m}$ operation: a finite-element re-evaluation gave the waveguide group index and mode overlap at the new wavelength and confirmed the resonator bend radius, $R = 231.406\text{ }\mu\text{m}$, as self-consistent, yielding a predicted scattering-only loss budget across the platform's four characterized widths. A more detailed benchmarking of the sidewall scattering contribution against measurement, including a discrepancy this comparison uncovers, is discussed in Chapter 3.

The two design projects' immediate next steps are stated explicitly in their respective chapters and left for the group to carry forward: fabrication and optical characterization of the Fabry-Perot devices, together with the ongoing bus-to-cavity coupling simulations, for the microresonator project (Chapter 2); and a measurement of the absorption contribution at $1\text{ }\mu\text{m}$, needed to complete the loss budget at that wavelength, for the scattering-loss project (Chapter 3). The three technical notes (Chapter 4) and the three software tools (Chapter 5) are in active use by the group and are maintained through the same documentation and version-control channels as the rest of the lab's shared infrastructure.